

Cryo-EM structure of the CBC-ALYREF complex

Reviewed Preprint

v2 • August 29, 2024

Revised by authors

Reviewed Preprint

v1 • October 27, 2023

Bradley P Clarke, Alexia E Angelos, Menghan Mei, Pate S Hill, Yihu Xie , Yi Ren 

Department of Biochemistry, Vanderbilt University School of Medicine Basic Sciences, Nashville, TN, USA • Center for Structural Biology, Vanderbilt University School of Medicine Basic Sciences, Nashville, TN, USA

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

Abstract

In eukaryotes, RNAs transcribed by RNA Pol II are modified at the 5' end with a 7-methylguanosine (m^7G) cap, which is recognized by the nuclear cap binding complex (CBC). The CBC plays multiple important roles in mRNA metabolism including transcription, splicing, polyadenylation, and export. It promotes mRNA export through direct interaction with a key mRNA export factor, ALYREF, which in turn links the TRanscription and EXport (TREX) complex to the 5' end of mRNA. However, the molecular mechanism for CBC mediated recruitment of the mRNA export machinery is not well understood. Here, we present the first structure of the CBC in complex with an mRNA export factor, ALYREF. The cryo-EM structure of CBC-ALYREF reveals that the RRM domain of ALYREF makes direct contact with both the NCBP1 and NCBP2 subunits of the CBC. Comparing CBC-ALYREF with other cellular complexes containing CBC and/or ALYREF components provides insights into the coordinated events during mRNA transcription, splicing, and export.

eLife assessment

This **important** study reports the cryo-electron microscopy structure of a multi-protein complex that recognizes the 5'-end cap of mRNAs and plays a critical role in mRNA export. The structural and biochemical analyses in this study provide **convincing** evidence to support the major claims of the authors, with the inclusion of more functional characterizations in cell-based systems having corroborated the claims further and thus strengthening the study. This paper would be of interest to structural biologists and RNA biologists working on mRNA metabolism.

<https://doi.org/10.7554/eLife.91432.2.sa4>

Introduction

The nuclear cap binding complex (CBC) binds to the m^7G cap of RNAs transcribed by RNA pol II. It is comprised of NCBP1 (also known as CBP80) and NCBP2 (also known as CBP20). This heterodimeric CBC can form a variety of interactions with different proteins to promote mRNA processing and influence the fate of the transcript. As such, the CBC regulates gene expression at multiple levels, ranging from transcription and splicing to nuclear export and translation^{1,2,3}.

The CBC is one of a myriad of protein factors that associate with newly synthesized transcripts. These factors package mRNAs into compacted ribonucleoprotein particles (mRNPs). While the overall structural arrangement of mRNPs is not known, factors such as serine/arginine-rich (SR) proteins are suggested to contribute to mRNP compaction. Prior to export, mRNPs acquire the export receptor NXF1-NXT1 to gain access to the nuclear pore complex (NPC)³. NXF1-NXT1 can interact with the FG repeats of the nucleoporin proteins in the NPC to mediate mRNA export^{4,5}. Intriguingly, early electron microscopic studies on the Balbiani ring mRNPs in *C. tentans* showed that mRNPs translocate through the NPC in a 5' to 3' direction⁶. Recent single molecule work on human mRNPs also suggests mRNA is exported 5' end first⁷.

The CBC promotes nuclear mRNA export through its interaction with a key mRNA export factor ALYREF. ALYREF in turn binds to DDX39B (also known as UAP56) and by extension, the entire TRanscription-EXport (TREX) complex at the 5' end of the RNA⁸. The TREX complex is conserved from yeast to humans^{9–13}. The human TREX complex is composed of THOC1, 2, 3, 5, 6, 7, and the DEAD-box helicase DDX39B¹⁴. Interestingly, ALYREF was also identified as the THOC4 subunit of the TREX complex, indicating that the function of ALYREF is tightly integrated with the TREX complex. ALYREF and its yeast ortholog Yra1 contain UBMs (UAP56-binding motifs) that mediate the interaction with DDX39B and Sub2, respectively^{10,15–17}. The yeast CBC was also shown to facilitate the recruitment of Yra1 onto nascent RNA¹⁸. ALYREF and TREX play central roles in mRNA export through direct interactions with various factors including the export receptor NXF1-NXT1^{19–21}. Their association with the 5' cap of mRNAs in particular is a key step in the export process. However, thus far the molecular mechanism underlying how ALYREF bridges the CBC and TREX remains unclear.

ALYREF and the associated TREX complex are not only required for cellular mRNA export but also can be hijacked for nuclear export of some viral mRNAs, such as those of herpes viruses. Two well studied examples are Herpes simplex virus (HSV-1) ICP27 and Herpesvirus saimiri (HVS) ORF57^{22,23}. Both HSV-1 ICP27 and HVS ORF57 directly target the host ALYREF protein^{24,25}. Structural studies show that they recognize overlapping surfaces on the RRM domain of ALYREF. How these viral factors affect host CBC-ALYREF interaction and function is not known.

The highly integrated nuclear mRNA processing and mRNP packaging also require the actions of multi-functional splicing factors, such as the SR protein SRSF1 and the exon junction complex (EJC). SRSF1 couples transcription, splicing, and export through direct interactions with the CBC, spliceosome, NXF1-NXT1, and RNA^{26–28}. Of note, ALYREF also has binding activities to these factors. Therefore, the functions of ALYREF and SRSF1 in mRNA processing and export are likely interconnected. In higher eukaryotes, the EJC is deposited 20–24 nucleotides upstream of the exon junctions during splicing^{29–31}. The EJC was initially shown to associate with ALYREF through a WxHD motif in the N-terminal unstructured region of ALYREF³². More recently, multiple binding interfaces were shown between ALYREF and the EJC¹². How the CBC-ALYREF connection affects the function of EJC-ALYREF remains elusive.

To better understand the multiple functions of ALYREF and the CBC in RNA metabolism, we carried out structural and biochemical studies. We present cryo-EM structures of the CBC and CBC-ALYREF complexes at 3.38 Å and 3.22 Å resolution, respectively. The CBC-ALYREF structure reveals that both the NCBP1 and NCBP2 subunits of the CBC interact with ALYREF. HSV-1 ICP27 and HVS ORF57 target the binding interface between the RRM domain of ALYREF and the CBC. We suggest that these viruses not only hijack host pathways to export their own RNA but could also inhibit host RNA metabolism through their interactions with ALYREF. Structural overlay of CBC-ALYREF and EJC-ALYREF reveals that both the CBC and the EJC bind to the RRM domain of ALYREF in a mutually exclusive manner. This suggests that ALYREF's interaction with the EJC is favored after ALYREF dissociates from the CBC, or as an independent event.

Results and discussion

ALYREF directly interacts with the CBC

The recombinant human ALYREF protein was shown to interact with the CBC in RNase-treated nuclear extracts⁸. Here we used purified recombinant proteins to further investigate the molecular interactions between ALYREF and the CBC (**Figure 1**). It is well known that RS domain containing proteins, including ALYREF, exhibit low solubility and are prone to aggregation. The addition of glutamic acid and arginine to the buffer can increase protein solubility and stability³³. Indeed, with a 1:1 mixture of glutamic acid and arginine, we were able to purify a GST-tagged ALYREF (residues 1-183) protein containing the N-terminal region and the RRM domain of ALYREF (**Figure 1A**). This construct includes the WxHD motif (residues 87-90), whose mutation was shown to affect interaction with the CBC in an immunoprecipitation study³². GST pull-down assays showed that ALYREF directly interacts with the CBC (**Figure 1B**).

We next attempted to characterize the CBC-ALYREF interaction using cryo-EM. We found that human ALYREF is prone to aggregation in the absence of a GST tag. To obtain an untagged ALYREF protein amenable for structural studies, we tested the solubility of several ALYREF orthologs and found that mouse ALYREF2 (mALYREF2, residues 1-155) exhibited better solubility than the ALYREF construct utilized above. The conserved UBM motifs, the WxHD motif, and the RRM domain are nearly identical between ALYREF and mALYREF2 (Figure 1-figure supplement 1). Indeed, like ALYREF, mALYREF2 directly interacts with the CBC (**Figure 1B**). As the ALYREF/mALYREF2 interaction with the CBC is conserved and mALYREF2 exhibits better solubility, we focused on mALYREF2 in the cryo-EM investigations.

Cryo-EM structures of the CBC and CBC-ALYREF complexes

We collected cryo-EM data from the mixture of the CBC complex and mALYREF2 in the presence of the cap analog m⁷GpppG. The data yielded two maps that correspond to the CBC complex and the CBC-mALYREF2 complex, respectively. The CBC map was refined to an overall resolution of 3.38 Å (**Figure 2A**, Figure 2-figure supplement 1, 2 and 3, **Table 1**). The electron density of the cap analog is clearly visible bound to NCBP2 (Figure 2-figure supplement 3C). Compared to unliganded CBC, the cap analog induces significant rearrangements in both the N-terminal extension and the C-terminal tail of NCBP2 (**Figure 2B**) to form critical interactions. For example, the N-terminal extension (residues 16-28) swings toward the central globular domain of NCBP2 and positions the Y20 residue to sandwich the cap analog with Y43 of NCBP2. These conformational changes were also observed in the crystal structures of ligand bound CBC^{34,35}. Overall, the cryo-EM structure of the CBC determined here resembles the previously reported crystal structures of the liganded CBC, with root mean squared deviation (RMSD) of 0.81 Å and 0.71 Å for NCBP1 and NCBP2, respectively³⁴.

The CBC-mALYREF2 map was refined to an overall resolution of 3.22 Å (**Figure 2C**, Figure 2-figure supplement 1, 4 and 5, **Table 1**). The cap analog is bound to NCBP2 within the CBC-mALYREF2 complex (Figure 2-figure supplement 5C). The structure shows that the RRM domain of mALYREF2 binds to both NCBP1 and NCBP2 subunits (**Figure 2C** and **2D**). The N terminal region of mALYREF2 (residues 1 to 73) does not show traceable density and is possibly disordered. mALYREF binding does not induce a significant overall conformational change in CBC (**Figure 2D**, Figure 2-figure supplement 5D). Comparing CBC and CBC-mALYREF2, NCBP1 and NCBP2 have an RMSD of 0.32 Å and 0.30 Å, respectively. Locally, an NCBP1 loop in proximity to mALYREF2, formed by residues 38-45, becomes more ordered.

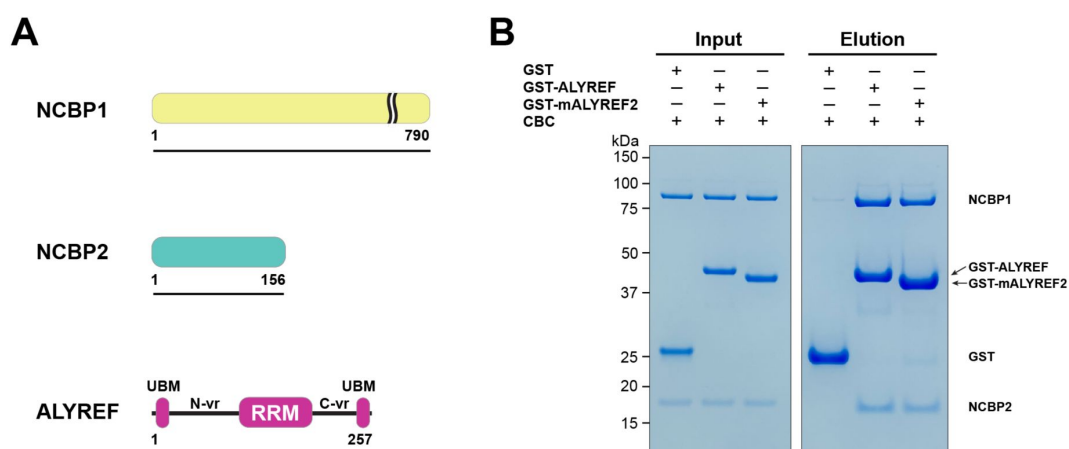


Figure 1.

ALYREF directly binds to CBC.

(A) Schematics of the NCBP1 and NCBP2 subunits of the human CBC and ALYREF. ALYREF contains a central RRM domain. The RRM domain is connected to two conserved motifs (UBMs) at both termini through variable regions (N-vr and C-vr). Protein constructs used in GST pull-down assays in panel B (NCBP1, residues 1-790; NCBP2, residues 1-156; ALYREF, residues 1-183) are indicated by black lines under the respective proteins. (B) ALYREF directly interacts with the CBC. In vitro GST pull-down assays were performed with purified recombinant human CBC and GST-tagged ALYREF or the corresponding construct of mouse ALYREF2 (mALYREF2, residues 1-155). Results are representative of three independent experiments.

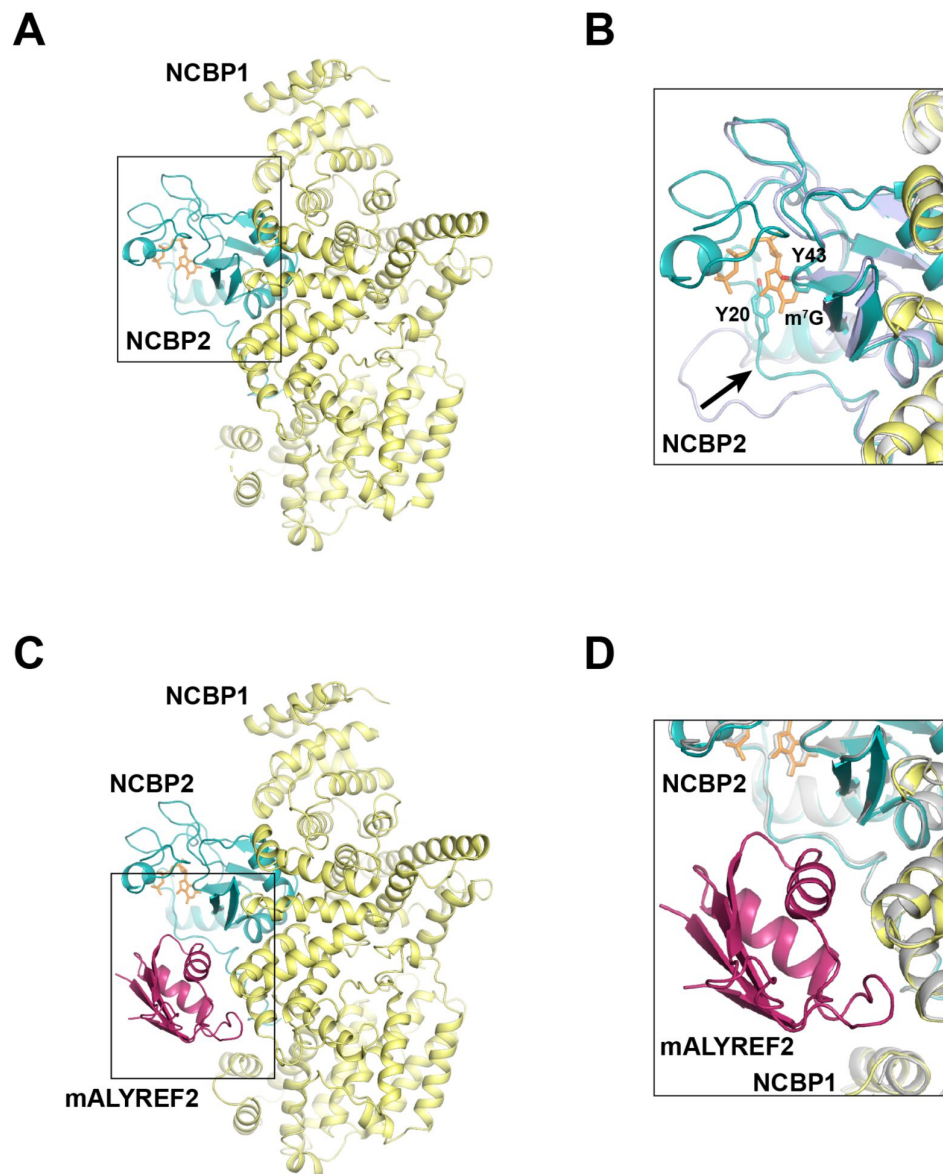


Figure 2.

Cryo-EM structures of CBC and CBC-mALYREF2

(A) Overall architecture of the CBC complex. NCBP1 and NCBP2 are colored in yellow and teal, respectively. The cap analog is shown as orange sticks. (B) Comparison of the CBC cryo-EM structure to the unliganded CBC crystal structure (PDB ID 1N54). The cryo-EM structure is colored as in (A). The unliganded CBC crystal structure is colored in light blue. Arrow indicates the conformational change in the N-terminal extension of NCBP2 upon cap analog binding. (C) Overall architecture of the CBC-mALYREF2 complex. NCBP1, NCBP2, and mALYREF2 are colored in yellow, teal, and red, respectively. The RRM domain of mALYREF2 binds to the CBC. (D) Comparison of the CBC-mALYREF2 and the CBC cryo-EM structures. CBC structure is colored in gray.

	CBC-mALYREF2 (EMDB EMD-40739) (PDB 8SRR)	CBC (EMDB EMD-40780) (PDB 8SUY)
Data collection and processing		
Microscope/Camera	Glacios/Falcon 4i	
Voltage (kV)	200	
Electron exposure (e-/Å ²)	52	
Defocus range (μm)	-1.0 to -2.0	
Pixel size (Å)	0.732	
Box size (pixels)	288	
Initial particle images (no.)	1,625,826	
Final particle images (no.)	241,915	78,039
Map resolution (masked, Å)	3.22	3.38
FSC threshold	0.143	0.143
Refinement		
Model resolution (masked, Å)	3.5	3.6
FSC threshold	0.5	0.5
Model composition		
Protein residues	975	895
Ligands	1	1
B factors (Å ²)		
Protein	162.9	180.1
Ligand	157.5	163.2
R.m.s. deviations		
Bond lengths (Å)	0.003	0.003
Bond angles (°)	0.417	0.413
Validation		
MolProbity score	1.34	1.48
Clashscore	6.24	7.37
Poor rotamers (%)	0.7	0.9
Ramachandran plot		
Favored (%)	98.0	97.6
Allowed (%)	2.0	2.4
Disallowed (%)	0.0	0.0

Table 1.

Cryo-EM data collection, refinement and validation statistics

CBC-ALYREF interfaces

The interfaces between mALYREF2 and the CBC involve 16, 18, and 6 residues of mALYREF2, NCBP1, and NCBP2, respectively (**Figure 3** [↗](#)). The RRM domain of mALYREF2 assumes a canonical $\beta 1\alpha 1\beta 2\beta 3\alpha 2\beta 4$ topology (**Figure 3A** [↗](#)), forming an α -helical surface and a β -sheet surface. The α -helical surface recognizes the CBC through extensive hydrophilic and hydrophobic interactions (**Figure 3B** [↗](#), Figure 3-figure supplement 1A and 1B). The $\alpha 1$ helix of mALYREF2 is enriched with acidic residues and makes key hydrophilic interactions with NCBP1. For example, E97 forms salt bridges with K330 and K381 of NCBP1. Y135 on the $\alpha 2$ helix of mALYREF2 makes a hydrogen bond with K330 of NCBP1. In addition, the loop between $\alpha 2$ and $\beta 4$ of mALYREF2 forms hydrophobic interactions with NCBP1. V138, P139, and L140 of mALYREF2 bind to a hydrophobic pocket on NCBP1 formed by A334, V337, and L382.

The interface between mALYREF2 and NCBP2 is near the m⁷G binding pocket (**Figure 3C** [↗](#), Figure 3-figure supplement 1C). The $\alpha 2$ helix of mALYREF2 contacts S13 and Y14 in the N-terminal extension of NCBP2. S13 and Y14 also directly interact with NCBP1 and are thought to enable the hinged motion of the N-terminal extension (residues 16-28) upon binding to the cap³⁵ [↗](#). In addition, the $\alpha 1$ helix of mALYREF2 is in proximity to the R105 and I110 residues of NCBP2. NCBP2 exhibits a positively charged groove extending from the cap binding site which is suggested to be an RNA binding site³⁴ [↗](#). Upon mALYREF2 binding, this groove is buried. Interestingly, mALYREF2 features a positively charged surface near the m⁷G site (**Figure 3D** [↗](#)). Conceivably, this positively charged surface on mALYREF2 could serve as an RNA binding site for the nucleotides following the cap.

Based on the CBC-mALYREF2 structure, we generated mutations (mut-1 and mut-2) on the RRM domain of human ALYREF (ALYREF-RRM, residues 103-183) to validate its interaction with the CBC. For ALYREF-RRM-mut-1 (Y166R/V169R/P170R), mutated residues are localized on the $\alpha 2$ - $\beta 4$ loop and correspond to residues Y135/V138/P139 in mALYREF2 (**Figure 3A** [↗](#), B). For ALYREF-RRM-mut-2 (E124R/E128R), mutated residues are localized on the $\alpha 1$ helix and correspond to residues E93/E97 in mALYREF2 (**Figure 3A** [↗](#), B). In agreement with the CBC-ALYREF structure, we found that the RRM domain of ALYREF directly interacts with the CBC, albeit with weaker interaction compared to ALYREF (residues 1-183) (**Figure 1B** [↗](#) and **4A** [↗](#)). The difference likely results from the WxHD motif (residues 87-90) localized in the N-terminal region of ALYREF. Evidence suggests that mutation of the WxHD motif reduces ALYREF's interaction with the CBC³² [↗](#). The WxHD motif may represent a second binding site for the CBC that remains to be characterized. Importantly, compared to the wild type protein, both ALYREF-mut-1 and mut-2 show reduced binding to the CBC (**Figure 4A** [↗](#)). Together, the mutagenesis studies validate the CBC-ALYREF interfaces observed in the structure.

The CBC-mALYREF2 structure reveals that the interaction between ALYREF and the CBC mainly involves the NCBP1 subunit (**Figure 3B** [↗](#) and **3C** [↗](#)). We further dissected the interaction between ALYREF and individual NCBP1 and NCBP2 subunits using GST pull-down assays. NCBP1 can be efficiently pulled down by GST-ALYREF, whereas NCBP2 did not show detectable interaction (**Figure 4B** [↗](#)). These results are consistent with the structural observations and indicate that NCBP1 is the major subunit of the CBC to interact with ALYREF.

CBC-ALYREF and 5' cap-dependent mRNP export

ALYREF recruits the mRNP export machinery TREX complex to the 5' end of mRNA through direct interactions with both the CBC and TREX^{8,10} [↗](#). The UBMs of ALYREF directly interact with the DDX39B component of the TREX complex¹⁰ [↗](#). The N-terminal UBM is included in the ALYREF construct used for our cryo-EM studies but did not show visible electron density. Thus, this UBM is likely exposed and available to interact with DDX39B, which further connects to the entire TREX complex (**Figure 5A** [↗](#)). Consistently, ALYREF, DDX39B, THOC1, and THOC2 are present in NCBP1

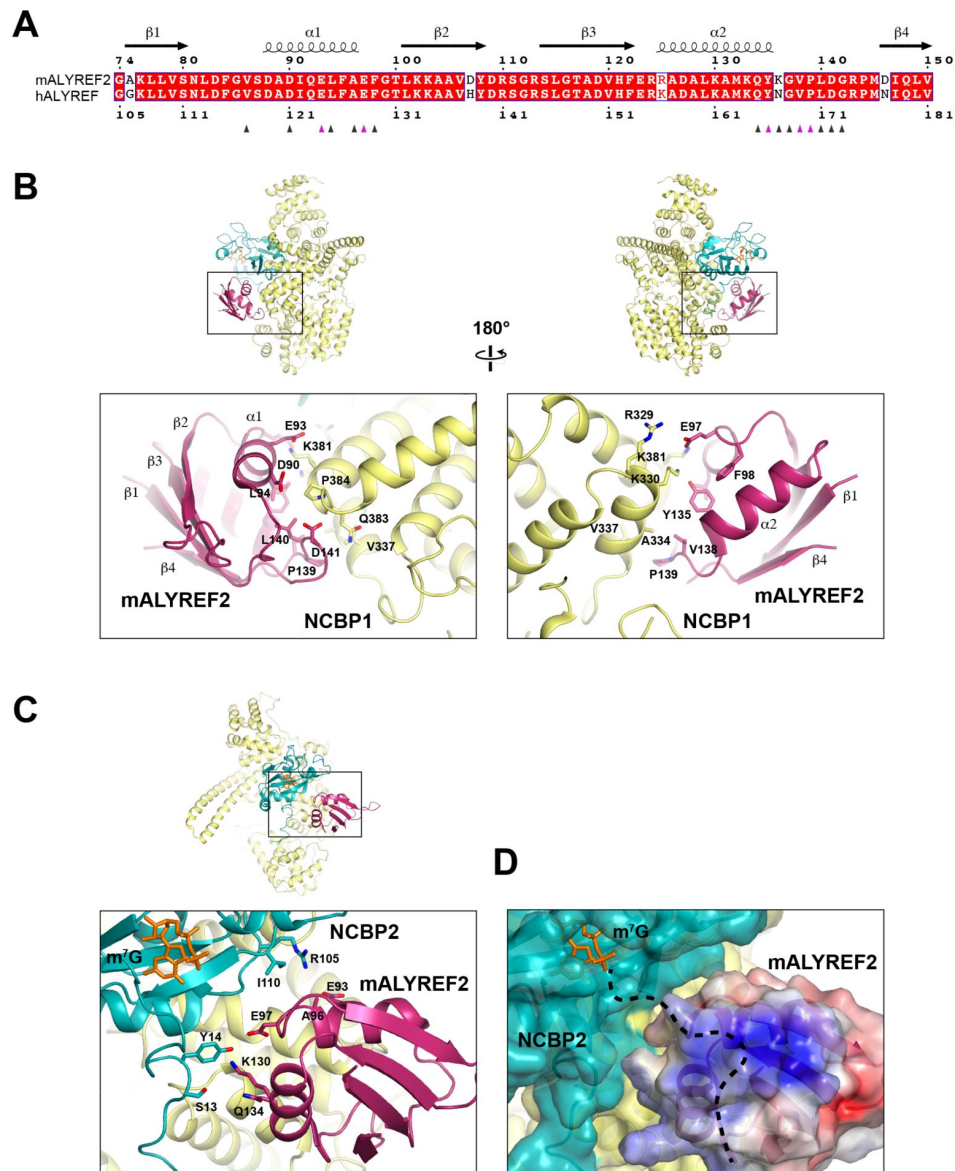


Figure 3.

ALYREF binds to both NCBP1 and NCBP2.

(A) Sequence alignment of the RRM domain of mALYREF2 and ALYREF⁶⁶. The residues that interface with the CBC are indicated by triangles below the sequence. The triangles colored in purple correspond to residues subjected to mutagenesis in **Figure 4**. (B) Details of the interaction between NCBP1 and mALYREF2. As a reference, an overall model is shown on the top to indicate the zoomed-in area. (C) Details of the interaction between NCBP2 and mALYREF2. (D) mALYREF2 features a positively charged surface near the cap analog bound to NCBP2. The surface of mALYREF2 is colored according to the electrostatic potential, ranging from red ($-5 k_B T/e$) to blue ($+5 k_B T/e$). Dotted line indicates a putative RNA binding path.

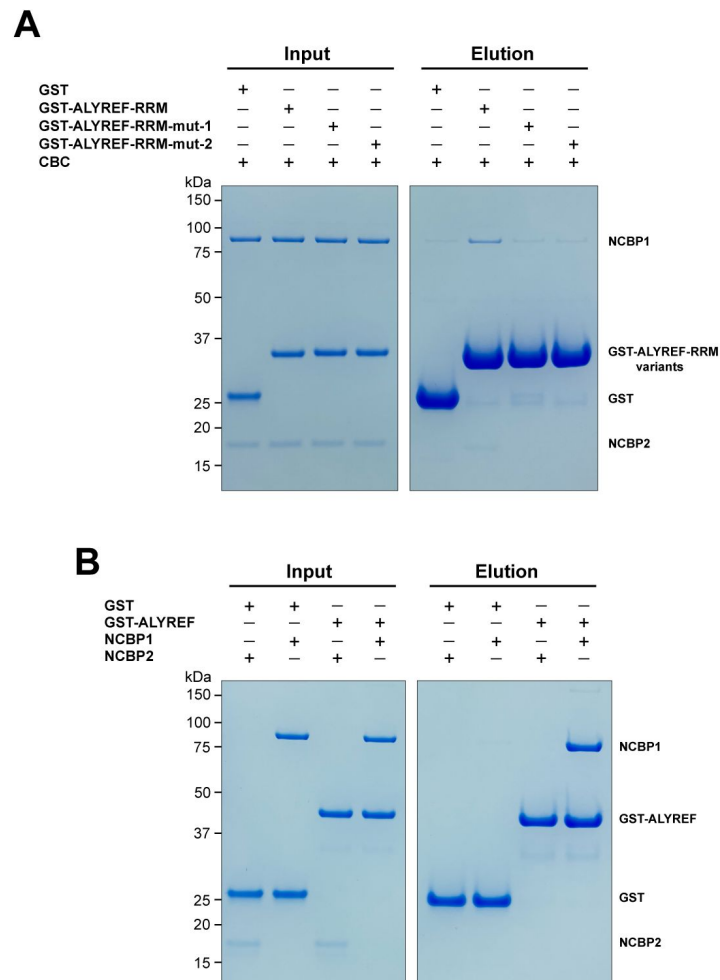


Figure 4.

Dissection of the ALYREF and CBC interfaces.

(A) Mutations of key interface residues on ALYREF reduced CBC binding. In vitro GST pull-down assays were performed with purified recombinant human CBC and GST-tagged ALYREF-RRM wild type or mutants (mut-1, Y166R/V169R/P170R; mut-2, E124R/E128R). (B) NCBP1 is sufficient to interact with ALYREF. In vitro GST pull-down assays were performed with purified recombinant GST-tagged ALYREF and individual CBC subunits. Results are representative of three independent experiments.

immunoprecipitations from RNase-treated HeLa cell nuclear extracts⁸. In yeast, mutually exclusive interactions were shown between Yra1 with Sub2 and the NXF1-NXT1 ortholog Mex67-Mtr2¹⁷. So, the ALYREF dependent NXF1-NXT1 loading on mRNA likely occurs after DDX39B dissociates from ALYREF. The CBC could also function as a landing pad for ALYREF as previously proposed³⁶. After recruitment to the 5' end of mRNA by the CBC, ALYREF could then transfer away from the 5' end, to other sites enriched with export factors and participate in different complexes located along the mRNA. In addition to the ALYREF-NXF1-NXT1 complex, some other ALYREF containing complexes could exist on the same mRNP, such as the complex of ALYREF/DDX39B/SARNP, which facilitates high order mRNP assembly^{37,38}.

The process of mRNP export, with the 5' end exiting first from the NPC, has been shown in both insect and human systems using electron microscopy and single molecule imaging techniques^{6,7}. Interestingly, in the latter study, several adjacent NPCs were found to engage in the export of the same mRNA³⁹. This observation is reminiscent of the gene gating hypothesis which suggested that transcriptionally active genes are physically tethered to the site of mRNA export at the NPC³⁹. Gated genes have been shown in yeast, worms, flies, and humans^{40,41}. For these gated genes, the 5' directionality of mRNA export could be primarily driven by the key placement of crucial RNA export factors at the 5' end of the gene as illustrated here (**Figure 5A**), and this localization of export factors could greatly increase the efficiency of co-transcriptional processing and export.

CBC-ALYREF and viral hijacking of host mRNA export pathway

HSV-1 ICP27 and HVS ORF57 hijack the host mRNA export pathway through interactions with ALYREF. The RRM domain of ALYREF is targeted by both HSV-1 ICP27 and HVS ORF57 with overlapping interfaces (**Figure 5B**, Figure 5-figure supplement 1). Structural comparison between CBC-ALYREF, ALYREF-ICP27, and ALYREF-ORF57 reveals that the interface between ALYREF's RRM domain with the CBC is not compatible with the ICP27/ORF57-ALYREF interactions (**Figure 5B**, Figure 5-figure supplement 1). In addition, in vivo data show that the ORF57 ortholog from Kaposi's Sarcoma-Associated Herpesvirus (KSHV) can still form a complex with ALYREF and the CBC²². So, it is likely that although ALYREF's RRM domain interface with the CBC could be disrupted by ORF57, ALYREF can still use the WxHD motif to interact with the CBC (**Figure 5C**). Using this strategy, a virus can hijack the host pathway while simultaneously disrupting host interactions and processes. It should also be noted that the CBC, ALYREF, ICP27, and ORF57 are all RNA binding proteins. In addition to the protein mediated interactions discussed above, RNA interactions should be considered, especially under the in vivo setting (**Figure 5C**). NXF1-NXT1 and DDX39B, the cellular ALYREF interacting proteins, are also hijacked by other factors from viruses. NXF1-NXT1 is targeted by influenza A virus NS1 protein⁴² and SARS-CoV-2 Nsp1 protein^{43,44}. DDX39B is targeted by influenza A virus NP protein^{45,46}. The molecular mechanisms revealed here and from previous studies pave the way for new useful targets in anti-viral therapeutics.

Functional interplay of CBC-ALYREF and mRNP export factors

Transcription, splicing, and export are all tightly linked processes. The CBC promotes splicing through its binding partners, such as SRSF1^{47,48}. The detailed molecular interaction between the CBC and SRSF1 is revealed in a human pre-B^{act-1} spliceosome structure²⁸. The NCBP2 subunit of the CBC is the major binding site for SRSF1. Structural overlay of this structure with the CBC-ALYREF structure shows no significant steric hindrance between SRSF1 and ALYREF (**Figure 6A**). However, whether the CBC-ALYREF-SRSF1 complex exits in vivo and how their functions might be coordinated require further studies. After splicing, SRSF1 is mainly deposited on exons^{49,50}. SRSF1 functions in mRNA export through interaction with the export receptor NXF1-NXT1^{26,51,52}. Of note, the interaction between both ALYREF and SR proteins with NXF1-NXT1 is regulated by phosphorylation. Only hypophosphorylated SR proteins can bind to

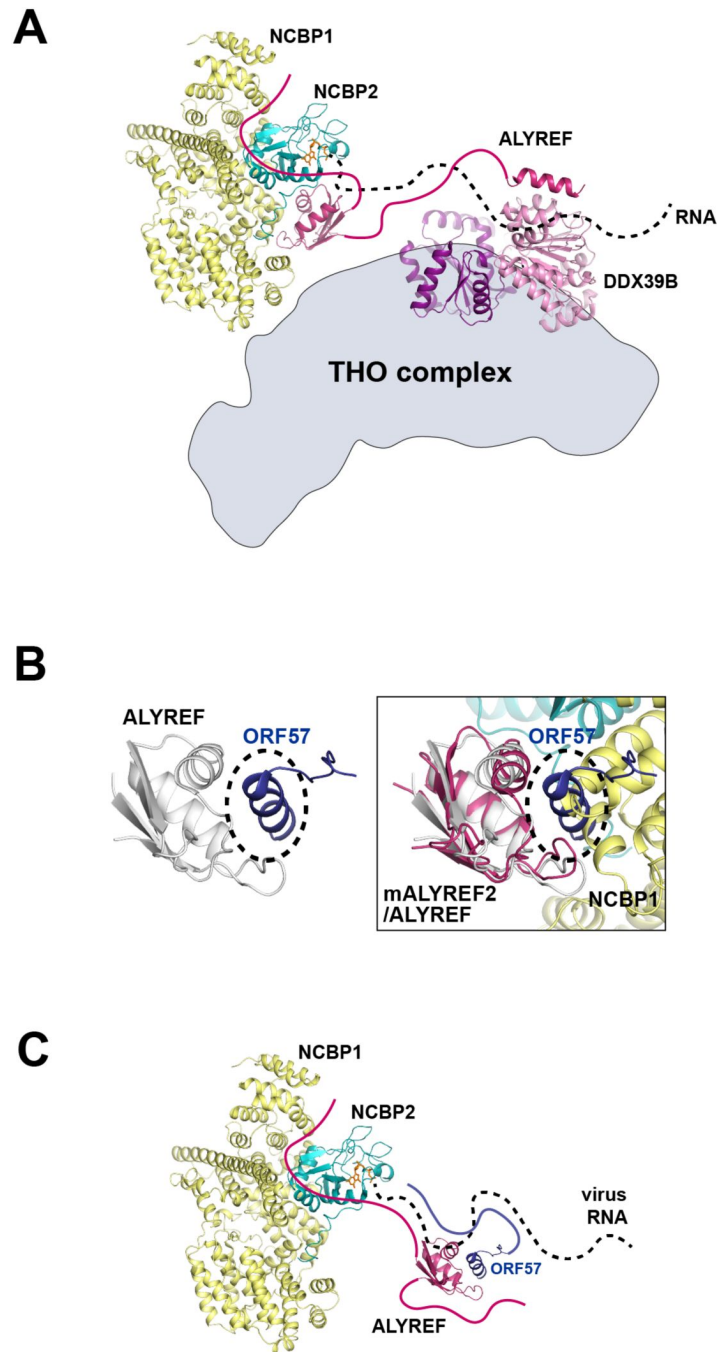


Figure 5.

Functional implications for ALYREF and CBC in 5' cap-dependent mRNP export.

(A) ALYREF links the TREX complex to the CBC. The RRM domain of ALYREF recognizes the CBC at the 5' end of mRNA. The UBM of ALYREF binds to the DDX39B subunit of TREX; their complex is represented using their yeast orthologs Yra1 and Sub2, respectively (PDB ID 5SUP). DDX39B in turn associates with the THO subcomplex of TREX. (B) HVS ORF57 binds to the RRM domain of ALYREF and interferes with the CBC-ALYREF interaction. (left) NMR structure of the ALYREF-ORF57 complex (PDB ID 2YKA). (right) The CBC-ALYREF structure is overlaid with the ALYREF-ORF57 structure. (C) Proposed model of viral mRNA export mediated by herpes viral ORF57 homologs. On viral transcripts, ALYREF associates with the CBC and ORF57 via the WxHD motif and the RRM domain, respectively. Both ALYREF and ORF57 feature RNA binding regions to form contacts with the RNA. ALYREF recruits the other TREX complex components to facilitate the nuclear export of viral mRNAs.

NXF1-NXT1 efficiently^{26,53,54}. Interestingly, SR proteins, such as Gbp2 and Hrb1 in *S. cerevisiae*, also interact directly with the TREX complex^{11,55}, suggesting coordinated actions of TREX and SR proteins in mRNA export.

Unlike the CBC and SRSF1 interfaces discussed above, the interaction between the ALYREF RRM domain and the CBC is incompatible with the interface between the ALYREF RRM domain and the EJC subunit MAGOH¹² (**Figure 6B**). Mutation of the ALYREF WxHD motif affects its interaction with both the CBC and the EJC subunit eIF4A3³². As both the WxHD motif and the RRM domain of ALYREF are mutually exclusive binding sites for the CBC and the EJC, the formation of the EJC-ALYREF complex likely happens after ALYREF dissociates from the CBC. It is also possible that the EJC-ALYREF interaction is independent of the CBC (**Figure 6C**). This possibility is supported by the report that ALYREF can be recruited to RNA by both CBC dependent and independent mechanisms⁵⁶. The resulting mRNP with multiple copies of ALYREF and NXF1-NXT1, each recruited through different mechanisms and at different sites on the mRNP (**Figure 6C**), could exhibit increased export efficiency.

Conclusion and perspectives

The CBC plays important roles in multiple steps of mRNA metabolism through interactions with a plethora of factors. Here we present the structural basis of the interaction between the CBC and a key mRNA export factor, ALYREF. The CBC-ALYREF structure reveals molecular insights into the ALYREF mediated recruitment of mRNA export machinery to the 5' end of nascent transcripts. We suggest working models for the coordinated events mediated by the CBC and ALYREF during splicing and mRNA export. Notably, both the CBC and ALYREF have been implicated in cancer. Mutations of the CBC residues interfacing with ALYREF, including K330N of NCBP1, R105C of NCBP2, and I110M of NCBP2, are found in several forms of cancers^{57–59}. These mutations could reduce ALYREF interaction and subsequently cause dysregulation of mRNA export and the processes that are coordinated by CBC and ALYREF. In addition, ALYREF has been shown to be frequently upregulated in various cancerous tissues⁶⁰. Determining whether dysregulation of the CBC-ALYREF interaction contributes to cancer pathogenesis, including characterization of the impact of cancer-associated mutations of the CBC, is an interesting area for future studies.

To date, it is unclear whether the functional connection between the CBC and ALYREF is conserved in yeast. In humans, recruitment of the mRNA export machinery is mediated by ALYREF in a CBC and EJC dependent manner^{8,32}. Our work and studies by others^{12,32} reveal that both the CBC and the EJC recognize ALYREF through the WxHD motif and the RRM domain of ALYREF. There are notable differences between yeast and humans regarding ALYREF function: 1) the EJC is not present in the budding yeast, *S. cerevisiae*, and 2) the yeast ortholog of ALYREF, Yra1, does not contain a WxHD motif. Nevertheless, a study conducting transcriptome-wide mapping of mRNA biogenesis factors showed that Yra1 is enriched at the 5' region of the mRNA⁶¹, raising the possibility that Yra1 localization could be mediated by the CBC. Further studies are required to reveal the mechanism of Yra1 in the context of the CBC and recruitment of the mRNA export machinery in yeast.

Materials and methods

Plasmids

Human NCBP1 (UniProt Q09161) was cloned into the pFastBac HTc vector with an N-terminal TEV cleavable His tag. Human NCBP2 (UniProt P52298) was cloned into a modified pFastBac1 vector containing an N-terminal TEV cleavable GST tag. Human ALYREF (UniProt Q86V81) and mouse

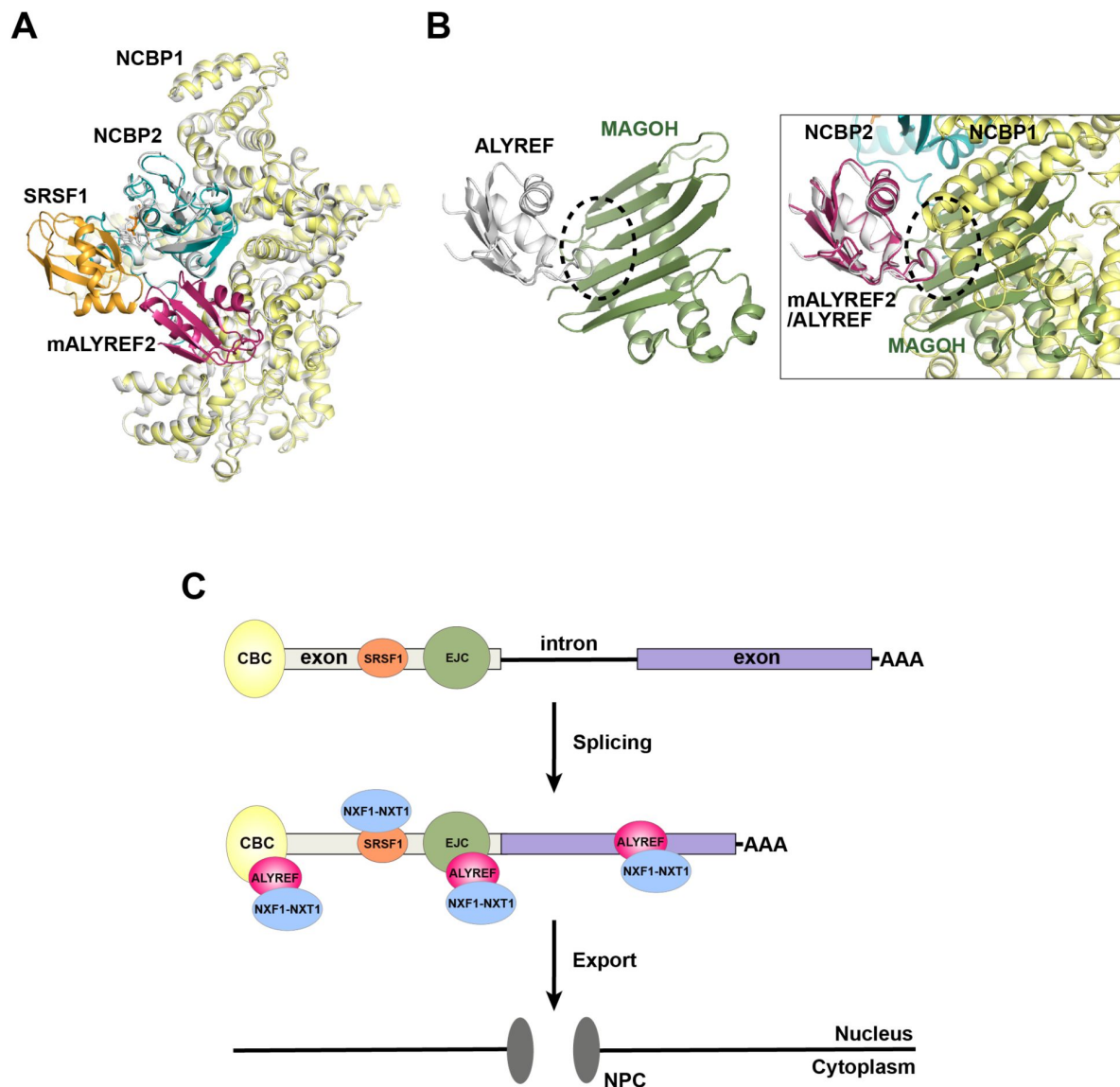


Figure 6.

ALYREF and CBC in splicing and export

(A) Overlay of the CBC-ALYREF structure with the CBC-SRSF1 structure (PDB ID 7ABG). CBC-ALYREF is colored as in **Figure 2B**. CBC-SRSF1 is colored with the CBC in white and SRSF1 in orange. (B) ALYREF interaction with the CBC and the EJC is mutually exclusive. (left) ALYREF binds to the MAGOH subunit of the EJC (PDB ID 7ZNI). (right) The CBC-ALYREF structure is overlaid with the ALYREF-MAGOH structure. (C) Proposed model of the mRNP export receptor NXF1-NXT1 recruitment by CBC-ALYREF and other factors during mRNA maturation.

ALYREF2 (UniProt Q4KL64) constructs were cloned into a modified pGST-4T-1 vector containing an N-terminal TEV cleavable GST tag.

Protein expression and purification

The NCBP1-NCBP2 complex was expressed in High-Five insect cells (Invitrogen) by coinfection of recombinant baculoviruses. Individual NCBP1 and NCBP2 subunits were expressed in High-Five insect cells infected with the respective recombinant baculovirus. High-Five cells were harvested 48 hrs after infection and lysed in a buffer containing 50 mM Tris pH 8.0, 300 mM NaCl, 0.2 mM AEBSF, 2 mg/L aprotinin, 1 mg/L pepstatin, 1 mg/L leupeptin, and 0.5 mM TCEP. Proteins were purified using Glutathione Sepharose 4B resin (Cytiva) for NCBP2 and NCBP1-NCBP2, or Ni Sepharose 6FF resin (Cytiva) for NCBP1 alone. The proteins were further purified on a mono Q column (Cytiva). The expression tags were removed by overnight incubation with GST-TEV (for NCBP2 and NCBP1-NCBP2) or His-TEV (for NCBP1) at 4 °C. The samples were passed through Glutathione Sepharose 4B resin (for NCBP2 and NCBP1-NCBP2) or Ni Sepharose 6FF resin (for NCBP1) to remove undigested protein and TEV. The proteins were further purified on a Superdex 200 column (Cytiva) in 10 mM Tris pH 8.0, 300 mM NaCl, and 0.5 mM TCEP.

GST-tagged ALYREF (residues 1-183), ALYREF-RRM (residues 102-183), ALYREF-RRM-mut-1 (Y166R/V169R/P170R), ALYREF-RRM-mut-2 (E124R/E128R), and mALYREF2 (residues 1-155) were expressed in *E. coli* Rosetta cells (Sigma-Aldrich). Protein expression was induced with 0.5 mM IPTG at 16°C overnight. Cells were lysed in a buffer containing 50 mM Tris, pH 8.0, 500 mM NaCl, 50 mM glutamic acid, 50 mM arginine, 0.5 mM TCEP, 0.2 mM AEBSF, and 2 mg/L aprotinin. The GST-tagged proteins were pulled down using Glutathione Sepharose 4B resin. GST-tagged ALYREF-RRM wild type and mutant proteins were purified on a Superdex 200 column in 10 mM Tris pH 8.0, 300 mM NaCl, and 5 mM DTT. GST-tagged ALYREF (residues 1-183) was purified by a cation exchange column (source 15S, Cytiva), followed by a Superdex 200 column equilibrated with 10 mM Tris, pH 8.0, 500 mM NaCl, 50 mM glutamic acid, 50 mM Arginine, and 0.5 mM TCEP. GST-tagged mALYREF2 (residues 1-155) was purified by an anion exchange column (Q Sepharose, Cytiva), followed by a cation exchange column (SP Sepharose, Cytiva). For untagged mALYREF2 used in cryo-EM studies, the GST tag was removed by overnight incubation with GST-TEV at 4 °C. Untagged mALYREF2 was further purified on a cation exchange HiTrap SP column (Cytiva), followed by a Superdex 200 column equilibrated with 10 mM Tris, pH 8.0, 150 mM NaCl, 50 mM glutamic acid, 50 mM Arginine, and 0.5 mM TCEP.

All purified proteins were flash frozen in liquid nitrogen, and stored at -80°C.

Cryo-EM sample preparation and data collection

NCBP1-NCBP2 at 1.2 μM was incubated with mALYREF2 (residues 1-155) at 3.6 μM in the presence of the 5' cap analog m⁷GpppG (NEB) at 500 μM at 4 °C for 0.5 hr. The sample was deposited on glow-discharged UltrAuFoil R 1.2/1.3 grids (Quantifoil). Grids were blotted for 6 s with a blotting force of 6 at 4 °C and 100% humidity and plunged into liquid ethane using a FEI Vitrobot Mark IV (Thermo Fisher). The data were collected with a Glacios Cryo-TEM (Thermo Fisher) equipped with a Falcon 4i detector (Thermo Fisher). Movies were collected with EPU at a magnification of 190,000x, corresponding to a calibrated pixel size of 0.732 Å/pixel. A total of 5858 movies recorded in EER format were collected with a defocus range from 1.0 μm to 2.0 μm. A full description of the cryo-EM data collection parameters can be found in [Table 1](#).

Cryo-EM image processing and Model building

Cryo-EM data were processed with cryoSPARC⁶². Movies in EER format were gain normalized, aligned and dose-weighted using patch motion correction, followed by patch CTF estimation. A subset of 100 micrographs was subjected to blob particle picking followed by 2D classification to obtain particles for Topaz training. 1,625,826 particles were picked by Topaz from the entire

dataset and were extracted with a box size of 288x288 pixels. 1,453,188 particles were retained after 2D classification. The particles were subjected to ab-initio model reconstruction and heterogeneous refinement, resulting in one good class from 637,185 particles which showed robust CBC densities as well as densities corresponding to mALYREF2 along with “junk” classes. Further heterogeneous refinement resulted in a set of 366,410 particles corresponding to the CBC-mALYREF2 complex and a set of 121,056 particles corresponding to the CBC complex. The CBC-mALYREF2 particles were subjected to another round of heterogeneous refinement and homogeneous refinement, which yielded a reconstruction from 241,915 particles with an overall resolution of 3.22 Å. Directional anisotropy of the CBC-mALYREF2 map was assessed using the 3DFSC server⁶³ (<https://3dfsc.salk.edu>) which indicates a sphericity of 0.946. The CBC particles were subjected to another round of heterogeneous refinement, homogeneous refinement, and non-uniform refinement, which yielded a reconstruction from 78,039 particles with an overall resolution of 3.38 Å. 3DFSC analysis of the CBC map indicates a sphericity of 0.962.

An initial model of CBC was obtained by docking AlphaFold models of NCBP1 (AF-Q09161-F1) and NCBP2 (AF-P52298-F1). An initial model of CBC-mALYREF2 was obtained by docking AlphaFold models of NCBP1, NCBP2, and mALYREF2 into the cryo-EM density map. The models were adjusted in Coot⁶⁴, followed by real-space refinement in Phenix⁶⁵. The final CBC model contains NCBP1, NCBP2, and the m⁷G moiety of m⁷GpppG. The final CBC-mALYREF2 model contains NCBP1, NCBP2, the RRM domain of mALYREF2, and the m⁷G moiety of m⁷GpppG. Figures were prepared using PyMOL (Molecular Graphics System, Schrodinger, LLC) and Chimera¹⁵.

GST pull-down

GST or GST-tagged proteins were pre-incubated with GST resin in binding buffer (10 mM Tris pH 8.0, 500 mM NaCl, 50 mM glutamic acid, 50 mM arginine, 0.5 mM TCEP) on ice for 0.5 hr and were mixed with gentle tapping every 3-5 minutes. The beads were then washed three times with 600 µl of buffer containing 10 mM Tris pH 8.0, 100 mM NaCl (**Figure 1B** and **4B**) or 50 mM NaCl (**Figure 4A**), and 0.5 mM TCEP. NCBP1, NCBP2, or the NCBP1-NCBP2 complex was incubated with the 5' cap analog m⁷GpppG at 20 µM, adjusted salt concentration to 100 mM NaCl (**Figure 1B** and **4B**) or 50 mM NaCl (**Figure 4A**), and then added to beads. The samples were incubated on ice for 0.5 hr and mixed with gentle tapping every 3-5 minutes. Beads were then washed three times with 600 µl of wash buffer (10 mM Tris pH 8.0, 50 mM NaCl, 0.5 mM TCEP) before bound proteins were eluted in 10 mM Tris pH 8.0, 500 mM NaCl (**Figure 1B** and **4B**) or 150 mM NaCl (**Figure 4A**), 25 mM Glutathione, and 0.5 mM TCEP. 6% (**Figure 1B** and **4B**) or 3% (**Figure 4A**) of the input and 60% of the eluted proteins were analyzed using Coomassie stained SDS-PAGE gels. The experiments were repeated three times independently.

Data availability

The cryo-EM structures of CBC-mALYREF2 (PDB ID 8SRR) and CBC (PDB ID 8SUY) have been deposited in the RCSB database. The cryo-EM maps of CBC-mALYREF2 (EMD-40739) and CBC (EMD-40780) have been deposited in the EMDB.

Acknowledgements

We thank Melissa Chambers, Scott Collier, and Mariam Haider at the Center for Structural Biology Cryo-EM Facility at Vanderbilt University for assistance in Cryo-EM data collection. We acknowledge the use of the Glacios cryo-TEM, which was acquired by NIH grant S10 OD030292-01.

Funding

This work was supported by NIH R35 GM133743 (Y.R.) and R01 AI184975 (Y.R.). B.P.C. was in part supported by NIH/NCI training grant T32CA119925.

Conflict of interest

The authors declare they have no conflict of interest.

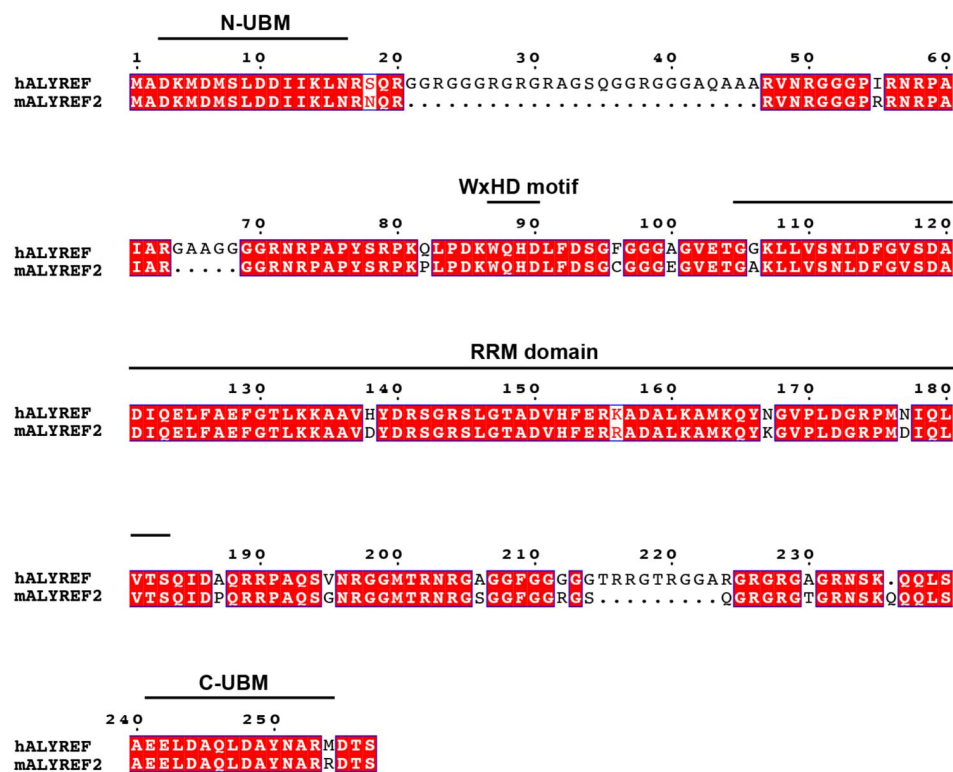


Figure 1 - figure supplement 1.

Human ALYREF and mouse ALYREF2 are conserved.

Sequence alignment of ALYREF and mALYREF2⁶⁶. The DDX39B binding motifs (N-UBM and C-UBM), the WxHD motif, and the RRM domain are indicated.

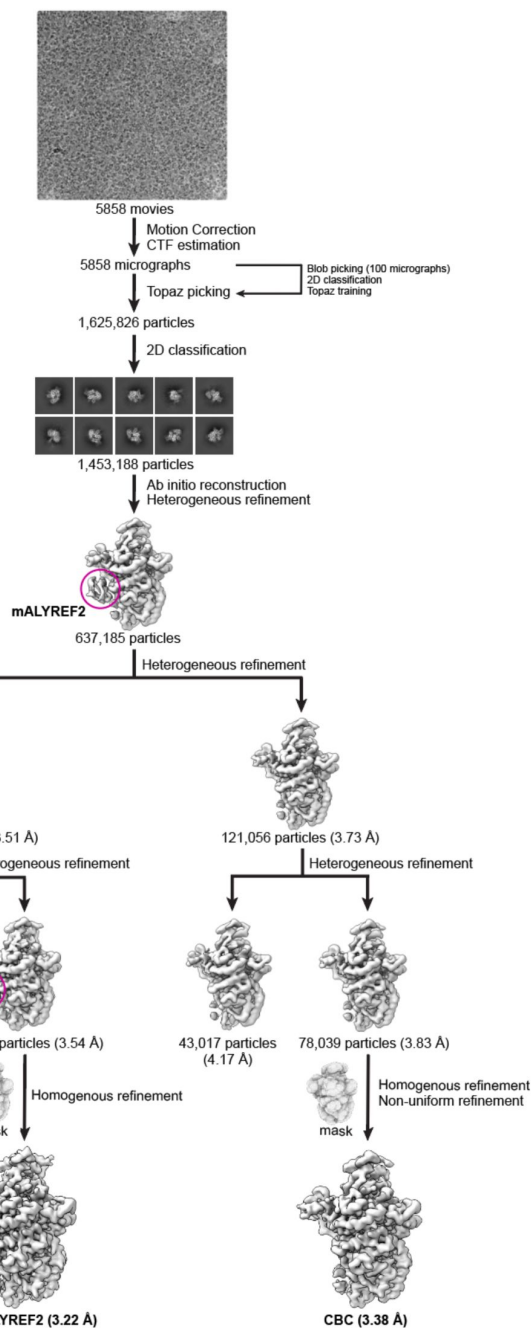


Figure 2 - figure supplement 1.

Workflow for cryo-EM data processing.

Data were processed in CryoSPARC. A set of 241,915 particles yielded a reconstruction of a CBC-mALYREF2 map at 3.22 Å resolution. A set of 78,039 particles yielded a reconstruction of a CBC map at 3.38 Å resolution.

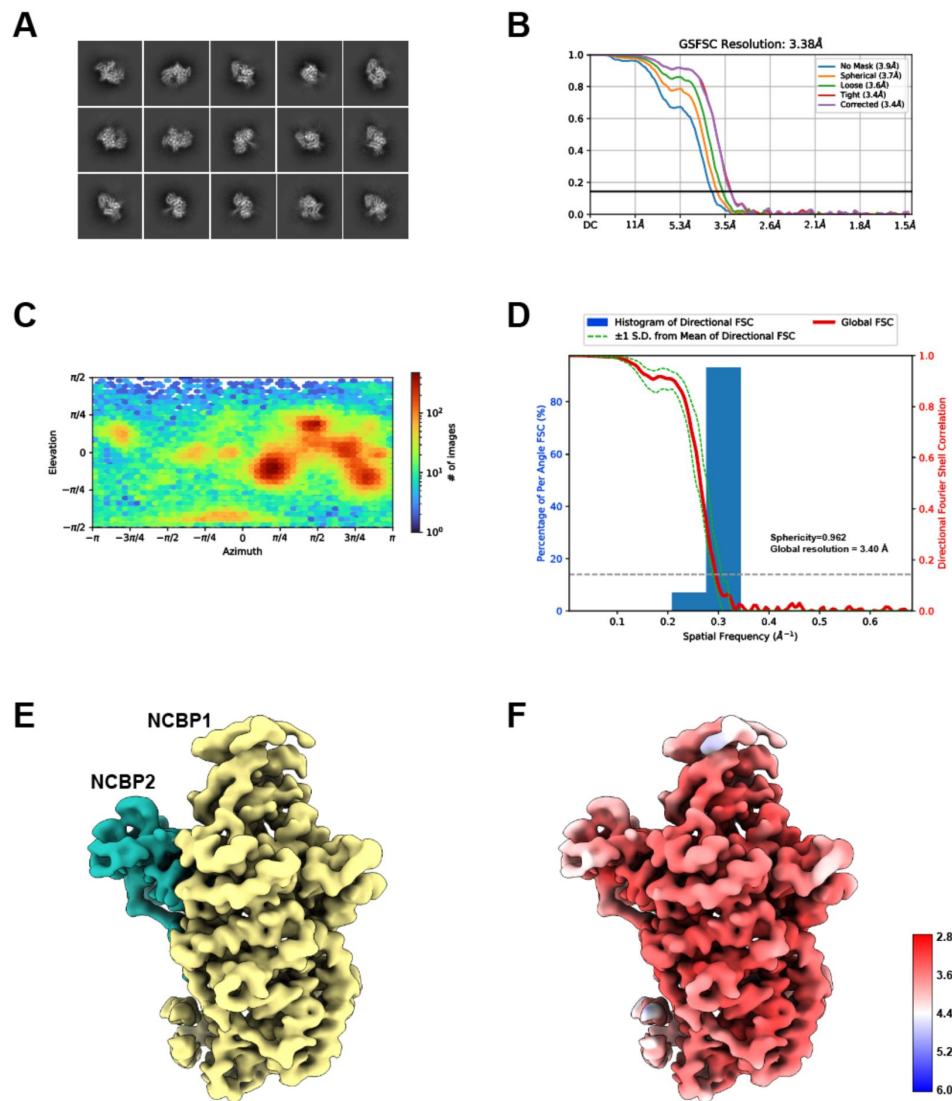


Figure 2 - figure supplement 2.

Cryo-EM reconstruction of the CBC.

(A) Selective 2D class averages from the final particle set of the CBC. (B) The gold-standard FSC curves of the final reconstruction of the CBC. (C) Angular distribution of the final reconstruction of the CBC computed in CryoSPARC. (D) 3DFSC analysis of the CBC reconstruction. The sphericity and global resolution values are noted within the graph window. (E) Cryo-EM map of the CBC complex. NCBP1 and NCBP2 are colored in yellow and teal, respectively. (F) Local resolution of the CBC cryo-EM map.

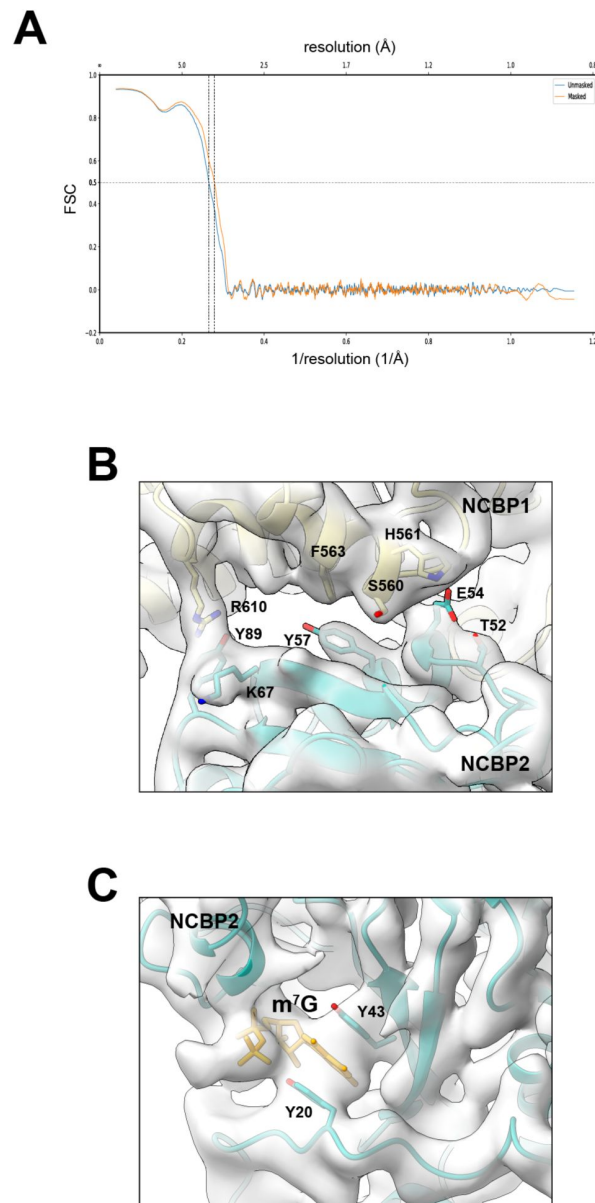


Figure 2 - figure supplement 3.

Structural model of the CBC.

(A) Map versus model FSC curves with or without mask calculated using Phenix. A resolution of 3.6 Å is estimated at FSC = 0.5.
 (B-C) Electron density map and structural model of CBC at the interface between NCBP1 and NCBP2 (B) and near m⁷G (C).

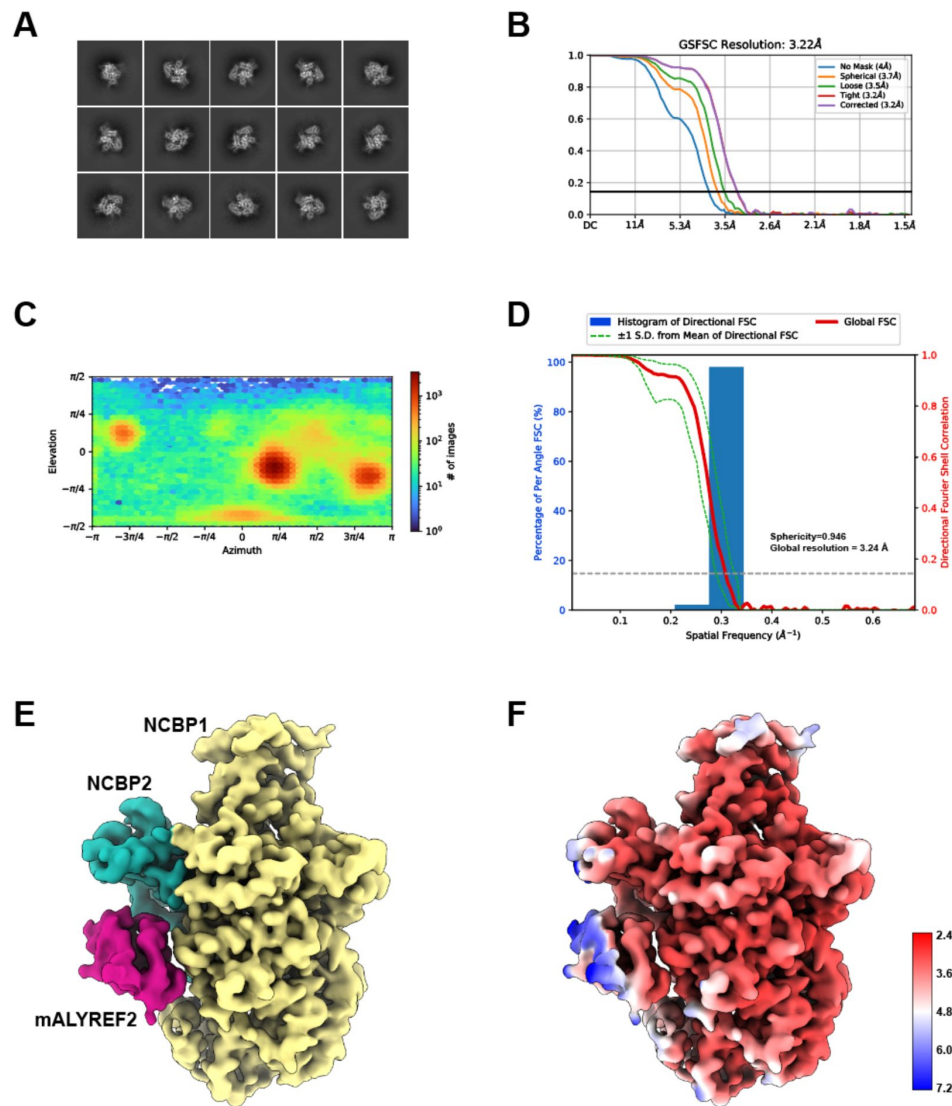


Figure 2 - figure supplement 4.

Cryo-EM reconstruction of the CBC-mALYREF2 complex.

(A) Selective 2D class averages from the final particle set of CBC-mALYREF2. (B) The gold-standard FSC curves of the final reconstruction of CBC-mALYREF2. (C) Angular distribution of the final reconstruction of CBC-mALYREF2 computed in CryoSPARC. (D) 3DFSC analysis of the CBC-mALYREF2 reconstruction. The sphericity and global resolution values are noted within the graph window. (E) Cryo-EM map of the CBC-mALYREF2 complex. NCBP1, NCBP2, and mALYREF2 are colored in yellow, teal, and red, respectively. (F) Local resolution of the CBC-mALYREF2 cryo-EM map.

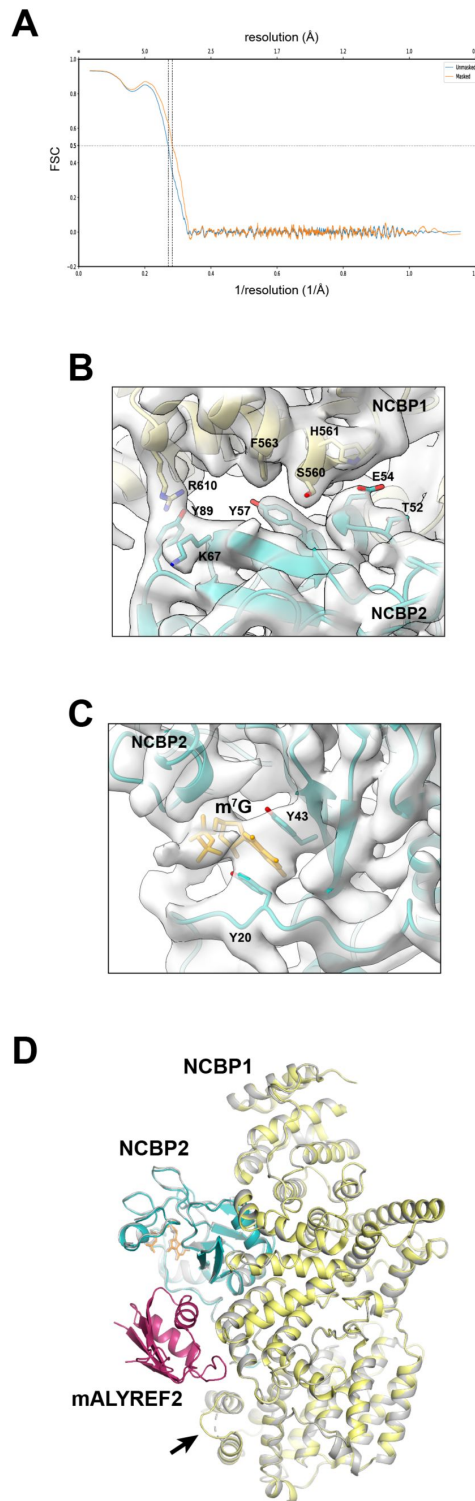


Figure 2 - figure supplement 5.

Structural model of the CBC-mALYREF2 complex.

(A) Map versus model FSC curves with or without mask calculated using Phenix. A resolution of 3.5 Å is estimated at FSC = 0.5. (B-C) Electron density map and structural model of CBC-mALYREF2 at the interface between NCBP1 and NCBP2 (B) and near m⁷G (C). (D) Comparison of the CBC-mALYREF2 (colored as in [Figure 2C](#)) and the CBC (gray) cryo-EM structures. A loop (residues 38-45) of NCBP1, indicated by the arrow, becomes ordered upon mALYREF2 binding.

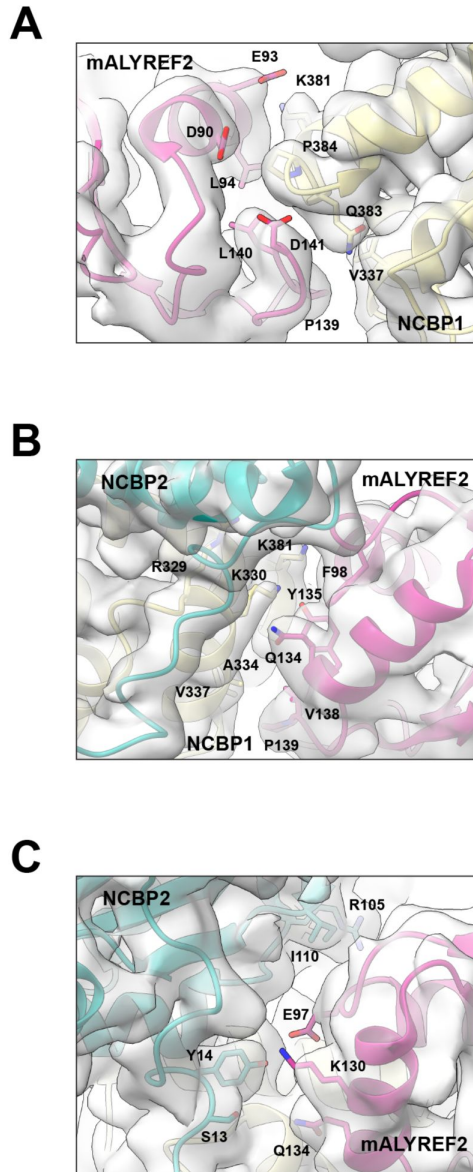


Figure 3 - figure supplement 1.

Electron density maps at the CBC-mALYREF2 interfaces.

Electron density map and structural model of CBC-mALYREF2 are shown in the same views as in **Figure 3B** [left panel \(A\)](#), **Figure 3B** [right panel \(B\)](#), and **Figure 3C** [\(C\)](#).

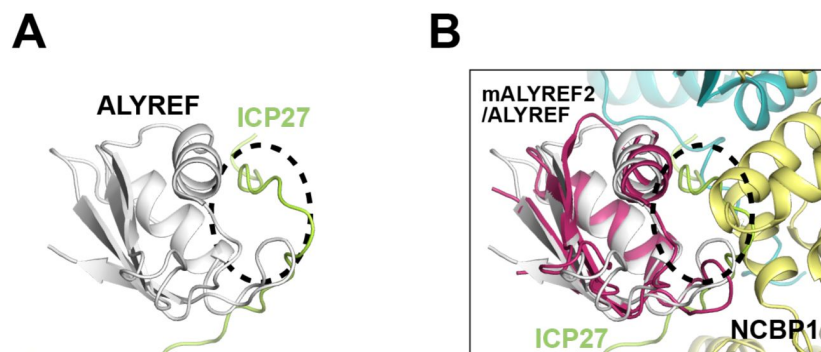


Figure 5 - figure supplement 1.

HSV-1 ICP27 binds to the RRM domain of ALYREF and interferes with the CBC-ALYREF interaction.

(A) NMR structure of the ALYREF-ICP27 complex (PDB ID 2KT5). (B) The CBC-ALYREF structure is overlaid with the ALYREF-ICP27 structure.

References

1. Gonatopoulos-Pournatzis T., Cowling V.H. (2014) **Cap-binding complex (CBC)** *Biochem J* **457**:231–42
2. Rambout X., Maquat L.E (2020) **The nuclear cap-binding complex as choreographer of gene transcription and pre-mRNA processing** *Genes Dev* **34**:1113–1127
3. Xie Y., Ren Y (2019) **Mechanisms of nuclear mRNA export: A structural perspective** *Traffic* **20**:829–840
4. Fribourg S., Braun I.C., Izaurrealde E., Conti E (2001) **Structural basis for the recognition of a nucleoporin FG repeat by the NTF2-like domain of the TAP/p15 mRNA nuclear export factor** *Mol Cell* **8**:645–56
5. Grant R.P., Neuhaus D., Stewart M (2003) **Structural basis for the interaction between the Tap/NXF1 UBA domain and FG nucleoporins at 1Å resolution** *J Mol Biol* **326**:849–58
6. Mehlin H., Daneholt B., Skoglund U (1992) **Translocation of a specific premessenger ribonucleoprotein particle through the nuclear pore studied with electron microscope tomography** *Cell* **69**:605–13
7. Ashkenazy-Titelman A., Atrash M.K., Boocholez A., Kinor N., Shav-Tal Y (2022) **RNA export through the nuclear pore complex is directional** *Nat Commun* **13**
8. Cheng H., et al. (2006) **Human mRNA export machinery recruited to the 5' end of mRNA** *Cell* **127**:1389–400
9. Strasser K., et al. (2002) **TREX is a conserved complex coupling transcription with messenger RNA export** *Nature* **417**:304–8
10. Ren Y., Schmiede P., Blobel G (2017) **Structural and biochemical analyses of the DEAD-box ATPase Sub2 in association with THO or Yra1** *Elife* **6**
11. Xie Y., et al. (2021) **Cryo-EM structure of the yeast TREX complex and coordination with the SR-like protein Gbp2** *Elife* **10**
12. Pacheco-Fiallos B., et al. (2023) **mRNA recognition and packaging by the human transcription-export complex** *Nature* **616**:828–835
13. Schuller S.K., et al. (2020) **Structural insights into the nucleic acid remodeling mechanisms of the yeast THO-Sub2 complex** *Elife* **9**
14. Masuda S., et al. (2005) **Recruitment of the human TREX complex to mRNA during splicing** *Genes Dev* **19**:1512–7
15. Luo M.L., et al. (2001) **Pre-mRNA splicing and mRNA export linked by direct interactions between UAP56 and Aly** *Nature* **413**:644–7

16. Chang C.T., et al. (2013) **Chtop is a component of the dynamic TREX mRNA export complex** *EMBO J* **32**:473–86
17. Strasser K., Hurt E (2001) **Splicing factor Sub2p is required for nuclear mRNA export through its interaction with Yra1p** *Nature* **413**:648–52
18. Sen R., et al. (2019) **Distinct Functions of the Cap-Binding Complex in Stimulation of Nuclear mRNA Export** *Mol Cell Biol* **39**
19. Viphakone N., et al. (2012) **TREX exposes the RNA-binding domain of Nxf1 to enable mRNA export** *Nat Commun* **3**
20. Strasser K., Hurt E (2000) **Yra1p, a conserved nuclear RNA-binding protein, interacts directly with Mex67p and is required for mRNA export** *EMBO J* **19**:410–20
21. Stutz F., et al. (2000) **REF, an evolutionary conserved family of hnRNP-like proteins, interacts with TAP/Mex67p and participates in mRNA nuclear export** *RNA* **6**:638–50
22. Boyne J.R., Colgan K.J., Whitehouse A (2008) **Recruitment of the complete hTREX complex is required for Kaposi's sarcoma-associated herpesvirus intronless mRNA nuclear export and virus replication** *PLoS Pathog* **4**
23. Sandri-Goldin R.M (2008) **The many roles of the regulatory protein ICP27 during herpes simplex virus infection** *Front Biosci* **13**:5241–56
24. Tunnicliffe R.B., et al. (2011) **Structural basis for the recognition of cellular mRNA export factor REF by herpes viral proteins HSV-1 ICP27 and HVS ORF57** *PLoS Pathog* **7**
25. Tunnicliffe R.B., Hautbergue G.M., Wilson S.A., Kalra P., Golovanov A.P (2014) **Competitive and cooperative interactions mediate RNA transfer from herpesvirus saimiri ORF57 to the mammalian export adaptor ALYREF** *PLoS Pathog* **10**
26. Huang Y., Yario T.A., Steitz J.A (2004) **A molecular link between SR protein dephosphorylation and mRNA export** *Proc Natl Acad Sci U S A* **101**:9666–70
27. Clery A., et al. (2013) **Isolated pseudo-RNA-recognition motifs of SR proteins can regulate splicing using a noncanonical mode of RNA recognition** *Proc Natl Acad Sci U S A* **110**:E2802–11
28. Townsend C., et al. (2020) **Mechanism of protein-guided folding of the active site U2/U6 RNA during spliceosome activation** *Science* **370**
29. Le Hir H., Izaurralde E., Maquat L.E., Moore M.J (2000) **The spliceosome deposits multiple proteins 20–24 nucleotides upstream of mRNA exon-exon junctions** *EMBO J* **19**:6860–9
30. Andersen C.B., et al. (2006) **Structure of the exon junction core complex with a trapped DEAD-box ATPase bound to RNA** *Science* **313**:1968–72
31. Bono F., Ebert J., Lorentzen E., Conti E (2006) **The crystal structure of the exon junction complex reveals how it maintains a stable grip on mRNA** *Cell* **126**:713–25
32. Gromadzka A.M., Steckelberg A.L., Singh K.K., Hofmann K., Gehring N.H (2016) **A short conserved motif in ALYREF directs cap- and EJC-dependent assembly of export complexes on spliced mRNAs** *Nucleic Acids Res* **44**:2348–61

33. Golovanov A.P., Hautbergue G.M., Wilson S.A., Lian L.Y (2004) **A simple method for improving protein solubility and long-term stability** *J Am Chem Soc* **126**:8933–9
34. Calero G., et al. (2002) **Structural basis of m7GpppG binding to the nuclear cap-binding protein complex** *Nat Struct Biol* **9**:912–7
35. Mazza C., Segref A., Mattaj I.W., Cusack S (2002) **Large-scale induced fit recognition of an m(7)GpppG cap analogue by the human nuclear cap-binding complex** *EMBO J* **21**:5548–57
36. Viphakone N., et al. (2019) **Co-transcriptional Loading of RNA Export Factors Shapes the Human Transcriptome** *Mol Cell* **75**:310–323
37. Dufu K., et al. (2010) **ATP is required for interactions between UAP56 and two conserved mRNA export proteins, Aly and CIP29, to assemble the TREX complex** *Genes Dev* **24**:2043–53
38. Xie Y., et al. (2023) **Structural basis for high-order complex of SARNP and DDX39B to facilitate mRNP assembly** *Cell Rep* **42**
39. Blobel G (1985) **Gene gating: a hypothesis** *Proc Natl Acad Sci U S A* **82**:8527–9
40. Burns L.T., Wente S.R (2014) **From hypothesis to mechanism: uncovering nuclear pore complex links to gene expression** *Mol Cell Biol* **34**:2114–20
41. Scholz B.A., et al. (2019) **WNT signaling and AHCTF1 promote oncogenic MYC expression through super-enhancer-mediated gene gating** *Nat Genet* **51**:1723–1731
42. Zhang K., et al. (2019) **Structural basis for influenza virus NS1 protein block of mRNA nuclear export** *Nat Microbiol* **4**:1671–1679
43. Zhang K., et al. (2021) **Nsp1 protein of SARS-CoV-2 disrupts the mRNA export machinery to inhibit host gene expression** *Sci Adv* **7**
44. Mei M., et al. (2024) **Inhibition of mRNA nuclear export promotes SARS-CoV-2 pathogenesis** *Proc Natl Acad Sci U S A* **121**
45. Momose F., et al. (2001) **Cellular splicing factor RAF-2p48/NPI-5/BAT1/UAP56 interacts with the influenza virus nucleoprotein and enhances viral RNA synthesis** *J Virol* **75**:1899–908
46. Morris A.K., et al. (2020) **Cellular mRNA export factor UAP56 recognizes nucleic acid binding site of influenza virus NP protein** *Biochem Biophys Res Commun* **525**:259–264
47. Lenasi T., Peterlin B.M., Barboric M (2011) **Cap-binding protein complex links pre-mRNA capping to transcription elongation and alternative splicing through positive transcription elongation factor b (P-TEFb)** *J Biol Chem* **286**:22758–68
48. Pabis M., et al. (2013) **The nuclear cap-binding complex interacts with the U4/U6.U5 tri-snRNP and promotes spliceosome assembly in mammalian cells** *RNA* **19**:1054–63
49. Das S., Krainer A.R (2014) **Emerging functions of SRSF1, splicing factor and oncoprotein, in RNA metabolism and cancer** *Mol Cancer Res* **12**:1195–204
50. Pandit S., et al. (2013) **Genome-wide analysis reveals SR protein cooperation and competition in regulated splicing** *Mol Cell* **50**:223–35

51. Reed R., Cheng H. (2005) **SR proteins and export of mRNA** *Curr Opin Cell Biol* **17**:269–73
52. Muller-McNicoll M., et al. (2016) **SR proteins are NXF1 adaptors that link alternative RNA processing to mRNA export** *Genes Dev* **30**:553–66
53. Okada M., Jang S.W., Ye K (2008) **Akt phosphorylation and nuclear phosphoinositide association mediate mRNA export and cell proliferation activities by ALY** *Proc Natl Acad Sci U S A* **105**:8649–54
54. Xie Y., et al. (2021) **Structure and activation mechanism of the yeast RNA Pol II CTD kinase CTDK-1 complex** *Proc Natl Acad Sci U S A* **118**
55. Hurt E., Luo M.J., Röther S., Reed R., Strässer K (2004) **Cotranscriptional recruitment of the serine-arginine-rich (SR)-like proteins Gbp2 and Hrb1 to nascent mRNA via the TREX complex** *Proc Natl Acad Sci U S A* **101**:1858–62
56. Nojima T., Hirose T., Kimura H., Hagiwara M (2007) **The interaction between cap-binding complex and RNA export factor is required for intronless mRNA export** *J Biol Chem* **282**:15645–51
57. Tate J.G., et al. (2019) **COSMIC: the Catalogue Of Somatic Mutations In Cancer** *Nucleic Acids Res* **47**:D941–D947
58. Heath A.P., et al. (2021) **The NCI Genomic Data Commons** *Nat Genet* **53**:257–262
59. Zhang J., et al. (2019) **The International Cancer Genome Consortium Data Portal** *Nat Biotechnol* **37**:367–369
60. Dominguez-Sanchez M.S., Saez C., Japon M.A., Aguilera A., Luna R (2011) **Differential expression of THOC1 and ALY mRNP biogenesis/export factors in human cancers** *BMC Cancer* **11**
61. Baejen C., et al. (2014) **Transcriptome maps of mRNP biogenesis factors define pre-mRNA recognition** *Mol Cell* **55**:745–57
62. Punjani A., Rubinstein J.L., Fleet D.J., Brubaker M.A (2017) **cryoSPARC: algorithms for rapid unsupervised cryo-EM structure determination** *Nat Methods* **14**:290–296
63. Tan Y.Z., et al. (2017) **Addressing preferred specimen orientation in single-particle cryo-EM through tilting** *Nat Methods* **14**:793–796
64. Emsley P., Lohkamp B., Scott W.G., Cowtan K. (2010) **Features and development of Coot** *Acta Crystallogr D Biol Crystallogr* **66**:486–501
65. Afonine P.V., et al. (2012) **Towards automated crystallographic structure refinement with phenix.refine** *Acta Crystallogr D Biol Crystallogr* **68**:352–67
66. Robert X., Gouet P (2014) **Deciphering key features in protein structures with the new ENDscript server** *Nucleic Acids Res* **42**:W320–4

Editors

Reviewing Editor

Yang Yang

Iowa State University

Senior Editor

Amy Andreotti

Iowa State University, Ames, United States of America

Reviewer #1 (Public Review):

Summary:

The authors use a combination of biochemistry and cryo-EM studies to explore a complex between the cap binding complex and an RNA binding protein, ALYREF, that coordinates mRNA processing and export.

Strengths:

The biochemistry and structural biology are supported by mutagenesis that tests the model in vitro. The structure provides new insight into how key events in RNA processing and export are likely to be coordinated.

Weaknesses:

The authors provide biochemical studies to confirm the interactions that they identify; however, they do not perform any studies to test these models in cells or explore the consequences for mRNA export from the nucleus. In fact, several of the amino acids that they identified in ALYREF that are critical for the interaction, as determined by their own biochemical studies are conserved in budding yeast Yra1 (residues E124/E128 are E/Q in budding yeast and residues Y135/V138/P139 are F/S/P), where the impact on poly(A) RNA export from the nucleus could be readily evaluated. The authors mention the potential for future studies in the manuscript, but they do not perform any analysis in this study that would explore the contributions of these new interactions.

<https://doi.org/10.7554/eLife.91432.2.sa3>

Reviewer #2 (Public Review):

Summary:

In this manuscript, Bradley and his colleagues represented the cryo-EM structure of the nuclear cap-binding complex (CBC) in complex with an mRNA export factor, ALYREF, providing a structural basis for understanding CBC regulating gene expression.

Strengths:

The authors successfully modeled the N-terminal region and the RRM domain of ALYREF (residues 1-183) within the CBC-ALYREF structure, which revealed that both the NCBP1 and NCBP2 subunits of the CBC interact with the RBM domain of ALYREF. Further mutagenesis and pull-down studies provided additional evidence to the observed CBC-ALYREF interface. Additionally, the authors engaged in a comprehensive discussion regarding other cellular complexes containing CBC and/or ALYREF components. They proposed potential models that elucidated coordinated events during mRNA maturation. This study provided structural

evidence to show how CBC effectively recruits mRNA export factor machinery, enhancing our understanding of CBC regulating gene expression during mRNA transcription, splicing, and export.

Weaknesses:

Absence of functional data to support the proposed models in this study.

<https://doi.org/10.7554/eLife.91432.2.sa2>

Reviewer #3 (Public Review):

Summary:

The authors carried out structural and biochemical studies to investigate the multiple functions of CBC and ALYREF in RNA metabolism.

Strengths:

For the structural study part, the authors successfully revealed how NCBP1 and NCBP2 subunits interact with mALYREF (residues 1-155). Their binding interface was then confirmed by biochemical assays (mutagenesis and pull-down assays) presented in this study.

Weaknesses:

The model derived from the cryo-EM structure will likely need to be validated in future functional studies.

<https://doi.org/10.7554/eLife.91432.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Reviews):

Summary:

The authors use a combination of biochemistry and cryo-EM studies to explore a complex between the cap-binding complex and an RNA binding protein, ALYREF, that coordinates mRNA processing and export.

Strengths:

The biochemistry and structural biology are supported by mutagenesis which tests the model in vitro. The structure provides new insight into how key events in RNA processing and export are likely to be coordinated.

Weaknesses:

The authors provide biochemical studies to confirm the interactions that they identify; however, they do not perform any studies to test these models in cells or explore the consequences of mRNA export from the nucleus. In fact, several of the amino acids that they identified in ALYREF that are critical for the interaction, as determined by their own biochemical studies, are conserved in budding yeast Yra1 (residues E124/E128 are E/Q in

budding yeast and residues Y135/V138/P139 are F/S/P), where the impact on poly(A) RNA export from the nucleus could be readily evaluated. The authors could at least mention this point as part of the implications and the need for future studies. No one seems to have yet targeted any of these conserved residues, so this would be a logical extension of the current work.

We thank the reviewer for the feedback on our work. ALYREF coordinates pre-mRNA processing and export through interactions with a plethora of mRNA biogenesis factors including the DDX39B subunit of the TREX complex, CBC, EJC, and 3' processing factors. ALYREF mediates the recruitment of the TREX complex on nascent transcripts which depends on its interactions with both CBC and EJC. Our work and studies by others indicate that ALYREF uses overlapping interfaces including both the N-terminal WxHD motif and the RRM domain to bind CBC and EJC. Thus, ALYREF mutants deficient in CBC interaction will also disrupt the ALYREF-EJC interaction and are not ideal for functional studies. In addition, the CBC plays important roles in multiple steps of mRNA metabolism through interactions with a plethora of factors, which often interact competitively with CBC. Identification of separation-of-function mutations on CBC or ALYREF that specifically disrupt their interaction but not other cellular complexes containing CBC or ALYREF would be an important future area to test the model in cells.

We appreciate the reviewer's insightful comments regarding yeast Yra1. Thus far, the physical and functional connection between Yra1 and CBC in yeast has not been demonstrated. There are major differences between yeast Yra1 and human ALYREF. Given the lack of an EJC in *S. cerevisiae*, it is unclear whether Yra1 acts in a similar manner as human ALYREF. In addition, Yra1 does not contain a WxHD motif in its N-terminal unstructured region, which is involved in CBC and EJC interactions in ALYREF. Characterization of the Yra1-CBC interaction will be an interesting future direction. We now include a discussion about yeast Yra1 in the newly added "Conclusion and perspectives" section.

Specific suggestions:

The authors could put their work in context by speculating how some of the amino acids that they identify as being critical for the interactions they identify could contribute to cancer. For example, they mention mutations of interacting residues in NCBP2 are associated with human cancers, pointing out that NCBP2 R105C amino acid substitution has been reported in colorectal cancer and the NCBP2 I110M mutation has been found in head and neck cancer. Do the authors speculate that these changes would decrease the interaction between NCBP2 and ALYREF and, if so, how would this contribute to cancer? They also mention that a K330N mutation in NCBP1 in human uterine corpus endometrial carcinoma, where Y135 on the $\alpha 2$ helix of mALYREF2 makes a hydrogen bond with K330 of NCBP1. How do they speculate loss of this interaction would contribute to cancer?

In the revised manuscript, we include a discussion about these CBC mutants found in human cancers in the "Conclusion and perspectives" section. We think some of these CBC mutants, such as NCBP-1 K330N, could reduce interaction with ALYREF. Compromised CBC-ALYREF interaction will affect the recruitment of the TREX complex on nascent transcripts and cause dysregulation of mRNA export. In addition, that could also change the partition of CBC and ALYREF in different cellular complexes and cause perturbation of various steps in mRNA biogenesis that are regulated by CBC and ALYREF. Thus far, it is unclear whether and how loss of the CBC-ALYREF interaction directly contributes to cancer. Our work and that of others provide molecular insights to test in future studies.

Reviewer #2 (Public Reviews):

Summary:

In this manuscript, Bradley and his colleagues represented the cryo-EM structure of the nuclear cap-binding complex (CBC) in complex with an mRNA export factor, ALYREF, providing a structural basis for understanding CBC regulating gene expression.

Strengths:

The authors successfully modeled the N-terminal region and the RRM domain of ALYREF (residues 1-183) within the CBC-ALYREF structure, which revealed that both the NCBP1 and NCBP2 subunits of the CBC interact with the RBM domain of ALYREF. Further mutagenesis and pull-down studies provided additional evidence to the observed CBC-ALYREF interface. Additionally, the authors engaged in a comprehensive discussion regarding other cellular complexes containing CBC and/or ALYREF components. They proposed potential models that elucidated coordinated events during mRNA maturation. This study provided good evidence to show how CBC effectively recruits mRNA export factor machinery, enhancing our understanding of CBC regulating gene expression during mRNA transcription, splicing, and export.

Weaknesses:

No in vivo or in vitro functional data to validate and support the structural observations and the proposed models in this study. Cryo-EM data processing and structural representation need to be strengthened.

We appreciate the reviewer's comments and suggestions. The fact that ALYREF uses highly overlapped binding interfaces for CBC and EJC interactions prevents us from a clear functional dissection of the ALYREF-CBC interaction using in vitro assays or in cells at the current stage. Please also see our response to Reviewer 1.

In this revised manuscript, we have reprocessed the cryo-EM data using a different strategy which yields significantly improved maps. We have made improvements to the presentation of the structural work based on the reviewer's specific comments.

Reviewer #3 (Public Reviews):

Summary:

The authors carried out structural and biochemical studies to investigate the multiple functions of CBC and ALYREF in RNA metabolism.

Strengths:

For the structural study part, the authors successfully revealed how NCBP1 and NCBP2 subunits interact with mALYREF (residues 1-155). Their binding interface was then confirmed by biochemical assays (mutagenesis and pull-down assays) presented in this study.

Weaknesses:

The authors did not provide functional data to support their proposed models. The authors should include more details regarding the workflow of their cryo-EM data processing in the figure.

We thank the reviewer for the comments. We completely agree that testing the proposed models in cells would be ideal. However, as we also respond to the other reviewers, functional studies are premature at the current stage because both ALYREF and CBC are components of many cellular complexes that regulate mRNA metabolism. Separation-of-function mutations on CBC or ALYREF first need to be identified in future studies for further investigation. Please also see our response to Reviewer 1.

As suggested by the reviewer, we have included more details of the cryo-EM workflow in this revised manuscript. We have also included various validation measures including 3DFSC analyses, map vs model FSC curves, and representative density maps at various protein-protein binding interfaces.

Recommendations for the Authors:

Reviewer #1 (Recommendations for the Authors):

Major points:

The authors should take advantage of Figure 1, which shows the domain structures of NCBP1, NCBP2, and ALYREF to indicate for the reader specifically which protein domains are included in the biochemical and structural analyses. In the current version of the manuscript, there is plenty of space to indicate below each domain structure precisely what regions are included.

In this revised manuscript, we have revised Figure 1A to indicate the protein constructs used in this work.

Although it is fine to combine the Results and Discussion, the authors should really offer a concluding paragraph to highlight the novel results from this study and put the results in context.

We thank the reviewer for the recommendation. We now include a “Conclusion and perspectives” section in this revised manuscript.

Minor comments:

Page 5, last sentence (and others) starts a sentence with the word "Since" when likely "As" which does not imply a time element to the phrase, is the correct word.

"Since the ALYREF/mALYREF2 interaction with the CBC is conserved and mALYREF2 exhibits better solubility, we focused on mALYREF2 in the cryo-EM investigations."

Would be more correct as: "As the ALYREF/mALYREF2 interaction with the CBC is conserved and mALYREF2 exhibits better solubility, we focused on mALYREF2 in the cryo-EM investigations."

We thank the reviewer for the comments. We have made the corrections.

The word 'data' is plural so the sentence at the bottom of p.9 that includes the phrase "...in vivo data shows.." should read "...in vivo data show.."

Corrected in the revised manuscript.

Reviewer #2 (Recommendations for the Authors):

Major points:

(1) The authors claimed the improved solubility of mouse ALYREF2 (mALYREF2, residues 1-155) compared to the previously employed ALYREF construct. However, human ALYREF has already been purified successfully for pull down assay, indicating soluble human ALYREF obtained, why not use human ALYREF directly? Please clarify.

Pull-down studies were performed with GST-tagged ALYREF. For cryo-EM studies, untagged ALYREF is preferred to avoid potential issues that may arise from the expression tag. However, untagged ALYREF is less soluble than GST-tagged ALYREF and is not amenable for structural studies. We have revised the text to clarify this point.

(2) The authors confirmed CBC-ALYREF interfaces through mutagenesis and pull-down assays in vitro. However, it would be more informative and interesting to include functional assays in vitro or/and in vivo with mutagenesis.

We completely concur with the reviewer that testing the proposed models in vitro and in vivo would be important. However, as we pointed out in our response to public reviews, the highly overlapped binding interfaces on ALYREF for CBC and EJC interactions pose a great challenge for functional studies. Furthermore, both ALYREF and CBC are multifunctional factors and interact with a number of partners. Ideally, separation-of-function mutants that specifically disrupt the CBC-ALYREF interaction but not others need to be identified in future studies in order to perform functional studies.

(3) About cryo-EM data processing and structural representation:

(1) In the description of the cryo-EM data processing, the authors claimed they did heterogeneous refinement, homogenous refinement, and then local refinement. This reviewer is puzzled by this process because the normal procedure should be non-uniform refinement following homogenous refinement. If the authors did not perform non-uniform refinement, they should do it because it would significantly improve the quality and resolution of cryo-EM maps. In addition, the right local refinement should include mask files and only show the density/map of the local region.

We thank the reviewer for the suggestions. In response to the reviewer's comment on the preferred orientation issue (point 5 below), we reprocessed the cryo-EM data and obtained significantly improved cryo-EM maps. In this revised manuscript, the CBC-mALYREF map was refined using homogeneous refinement; the CBC map was refined using homogenous refinement followed by non-uniform refinement. Refinement masks are included in Figure 2-figure supplement1.

(2) Further local refinements with signal subtraction should be performed to improve the density and resolution of mALYREF2.

We tested local refinement with or without signal subtraction using masks covering mALYREF2 and various regions of CBC. Unfortunately, this approach did not improve the density of mALYREF2. We suspect that the small size of mALYREF2 (77 residues for the RRM domain) and the intrinsic flexibility of CBC are the limiting factors in these attempts.

(3) Figures with cryoEM map showing the side chains of the residues on the CBC-mALYREF2 interface should be included to strengthen the claims. Authors could add the map to Figure 3b/c or present it as a supplementary figure.

We include new supplementary figures (Figure 3-figure supplement 1) to show the electron densities corresponding to the views in Figure 3B and 3C. Residues labeled in Figure 3B and 3C are shown in sticks in these supplementary figures.

(4) For cryo-EM data processing, the authors have omitted lots of important details. Could the authors elaborate on the data processing with more details in the corresponding Figure and Methods Sections? Only one ab-initio model from the picked good particles was displayed in the figure. Are there any other different conformations of 3D classes for the dataset? In addition, too few classes have been considered in 3D classification, more classes may give a class with better density and resolution.

We thank the reviewer for the comments. We have reprocessed the cryo-EM data. A major change is to use Topaz for particle picking. We now include more details for data processing in Figure 2-figure supplement 1 and the method section. The cryo-EM sample is relatively uniform. Ab-initio reconstruction and heterogeneous refinement yielded only one good class and the other classes are “junk” classes (omitted in the workflow figure). No major conformational changes were observed throughout the multiple rounds of heterogeneous refinement for both CBC and CBCmALYREF2. In this revised manuscript, we have been able to obtain significantly improved maps through the new data processing strategy employing Topaz as illustrated in Figure 2-figure supplement 1 to 5.

(5) Angular distribution plots should be included to show if there is a preferred orientation issue. Based on the presented maps in validation reports, there may exist a preferred orientation issue for the reported two cryo-EM maps. Detailed 3D-Histogram and directional FSC plots for all the cryo-EM maps using 3DFSC web server should be presented to show the overall qualities (<https://www.nature.com/articles/nmeth.4347> and <https://3dfsc.salk.edu/>).

We thank the reviewer for the recommendations. In response to the reviewer’s comment on the preferred orientation issue, we reprocessed the cryo-EM data. Topaz was used for particle picking instead of template picking. 3DFSC analyses indicate that the new CBC-mALYREF2 map has a sphericity of 0.946, which is a significant improvement from the previous map which has a sphericity of 0.815. Consistently, the maps presented in this revised manuscript show significantly improved densities. We now include angular distribution and 3DFSC analyses of the EM maps (Figure 2-figure supplement 2 and 4).

(6) Figures of model-to-map FSCs need to be present to demonstrate the quality of the models and the corresponding ones (model resolution when FSC=0.5) should also be included in Table 1. The accuracy of the model is important for structural explanations and description.

The model-to-map FSCs are now included in Figure 2-figure supplement 3A and 5A. The model resolutions of CBC-mALYREF2 and CBC are estimated to be 3.5 Å and 3.6 Å at an FSC of 0.5. These numbers are now included in Table 1.

(7) In addition, figures of local density maps with different regions of the models, showing side chains, are necessary and important to justify the claimed resolutions.

We now include density maps overlayed with residue side chains at various regions. For the CBCmALYREF2 map, density maps are shown at the mALYREF2-NCBP1 interfaces (Figure 3-figure supplement 1A and 1B), mALYREF2-NCBP2 interface (Figure 3-figure supplement 1C), NCBP1NCPB2 interface (Figure 2-figure supplement 5B), and the region near m7G (Figure 2-figure supplement 5C). For the CBC map, density maps are shown at the NCBP1-NCPB2 interface (Figure 2-figure supplement 3B) and the region near m7G (Figure 2-figure supplement 3C).

Minor points:

(1) A figure superimposing the models from the CBC-mALYREF2 amp and mALYREF2 alone map is necessary to present that there are no other CBC binding-induced conformational changes in CBC except the claimed by the authors. In addition, a figure showing the density of m7GpppG should be included as well.

Overlay of CBC and CBC-mALYREF2 models is now presented in Figure 2-figure supplement 3D. Comparing CBC and CBC-mALYREF2, NCBP1 and NCBP2 have a RMSD of 0.32 Å and 0.30 Å, respectively. The density maps near the M7G cap analog are shown in Figure 2-figure supplement 3C for CBC and Figure 2-figure supplement 5C for CBC-mALYREF2.

(2) Authors obtained the two maps from one dataset, so "we first determined" and "we next determined" (page 6) should be replaced with something like "One class of 3D cryo-EM map revealed" and "Another class of 3D cryo-EM map defined".

We have revised the text as suggested by the reviewer.

(3) In 'Abstract', 'a mRNA export factor' should be 'an mRNA export factor'.

Corrected in the revised manuscript.

(4) In 'Abstract', the final sentence 'Comparison of CBC- ALYREF to other CBC and ALYREF containing cellular complexes provides insights into the coordinated events during mRNA transcription, splicing, and export' doesn't read smoothly, I would suggest revising it to 'Comparing CBC-ALYREF with other cellular complexes containing CBC and/or ALYREF components provides insight into the coordinated events during mRNA transcription, splicing, and export.'

We thank the reviewer for the recommendation and have revised accordingly.

(5) In paragraph 'CBC-ALYREF and viral hijacking of host mRNA export pathway', line 6, the sentences preceding and following the term 'However' indicate a progressive or parallel relationship, rather than a transitional one. To enhance the coherence, I would suggest replacing 'However' with 'Furthermore' or 'In addition'.

Corrected in the revised manuscript.

(6) In both Figure 5 and Figure 6, the depicted models are proposed and constructed exclusively through the comparison of the CBC-partial ALYREF with other cellular complexes containing components of CBC and/or ALYREF, which need to be confirmed by more studies. To prevent potential confusion and misunderstandings, it is recommended to replace the term 'model' with 'proposed model'.

Corrected in the revised manuscript.

Reviewer #3 (Recommendations for the Authors):

Major points:

(1) In the Results and Discussion section, the authors mentioned "Recombinant human ALYREF protein was shown to interact with the CBC in RNase-treated nuclear extracts." However, they used mouse ALYREF for cryo-EM investigations. Can the authors include an explanation for this choice during the revision?

In our work, we used a mixture of glutamic acid and arginine to increase the solubility of GSTALYREF. For cryo-EM studies, we use untagged ALYREF to avoid potential issues that may arise from the expression tag. However, untagged ALYREF is less soluble than GST-tagged ALYREF and is not suitable for structural studies in standard buffers. We have made further clarification on this point in this revised manuscript.

(2) In the paragraph on "CBC-ALYREF interfaces", the authors stated "For example, E97 forms salt bridges with K330 and K381 of NCBP1. Y135 on the $\alpha 2$ helix of mALYREF2 makes a hydrogen bond with K330 of NCBP1. The importance of this interface between ALYREF and NCBP1 is highlighted by a K330N mutation found in human uterine corpus endometrial carcinoma." I fail to see a strong connection between their structural observations and previous findings regarding the role of a K330N mutation found in human uterine corpus endometrial carcinoma. The authors should add more words to thread these two parts.

In response to the reviewer's comment, we now move the discussion of these CBC mutants to the newly added "Conclusion and perspectives" section.

(3) The authors should include side chains of the residues in their figure of Local resolution estimation and FSC curves, especially when they are presenting the binding interface between two components.

We have now included density maps that are overlaid with structural models showing side chains of critical residues. These maps include the NCBP1-mALYREF2 interfaces (Figure 3-figure supplement 1A and 1B), NCBP2-mALYREF2 interface (Figure 3-figure supplement 1C), NCBP1NCBP2 interface (Figure 2-figure supplement 3B and 5B), and the m7G cap region (Figure 2-figure supplement 3C and 5C).

Minor points:

(1) Some grammatical mistakes need to be corrected. For example, it is "an mRNA" instead of "a mRNA".

Corrected in the revised manuscript.

(2) The authors can provide more information for the audience to know better about ALYREF when it first appears in the 5th line in the Abstract section. For example, "It promotes mRNA export through direct interaction with ALYREF, a key mRNA export factor, ...".

We have revised the sentence based on the reviewer's comment.

<https://doi.org/10.7554/eLife.91432.2.sa0>